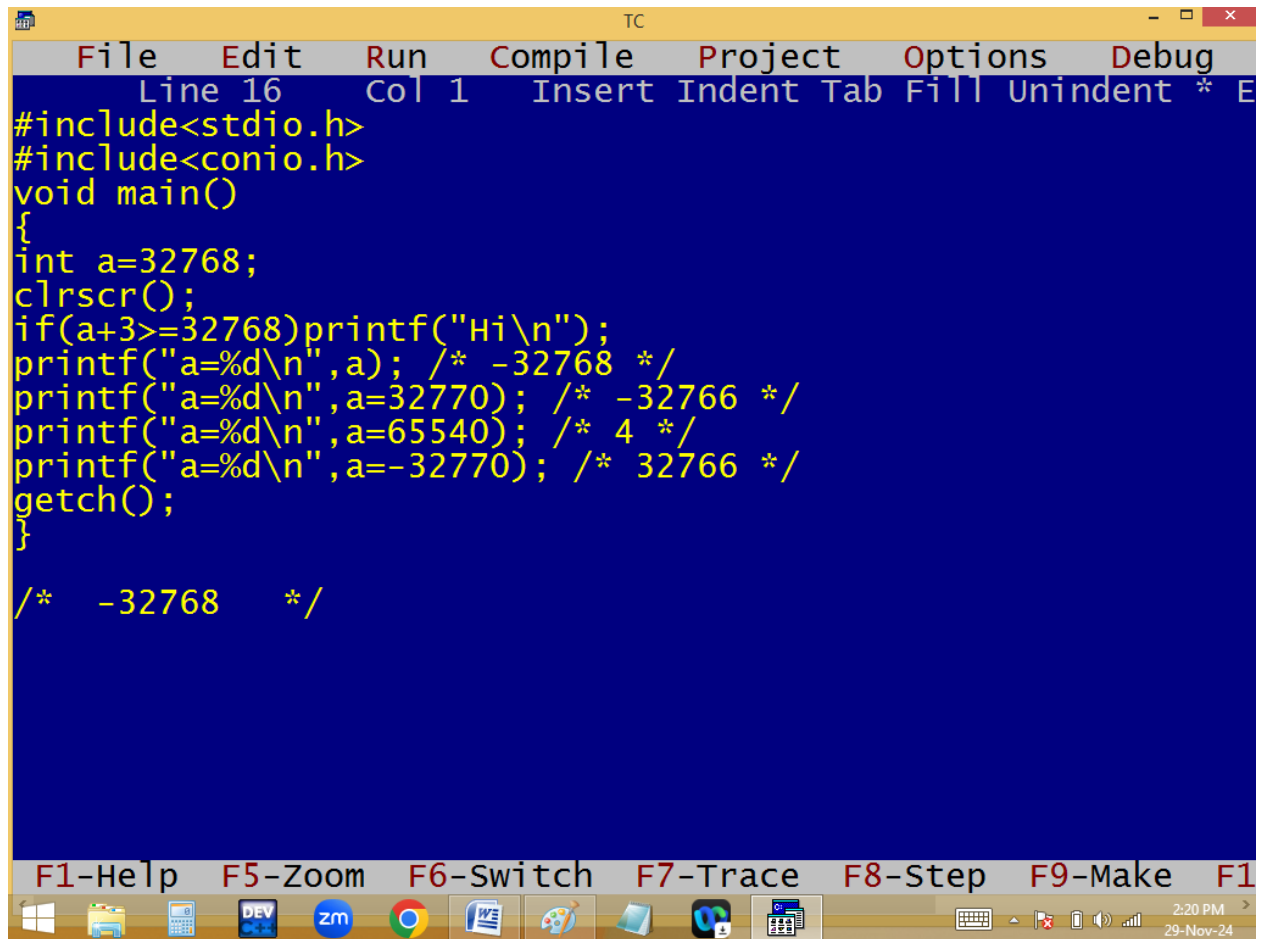




```
TC
File Edit Run Compile Project Options Debug
Line 16 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=32768;
clrscr();
if(a+3>=32768)printf("Hi\n");
printf("a=%d\n",a); /* -32768 */
printf("a=%d\n",a=32770); /* -32766 */
printf("a=%d\n",a=65540); /* 4 */
printf("a=%d\n",a=-32770); /* 32766 */
getch();
}

/* -32768 */

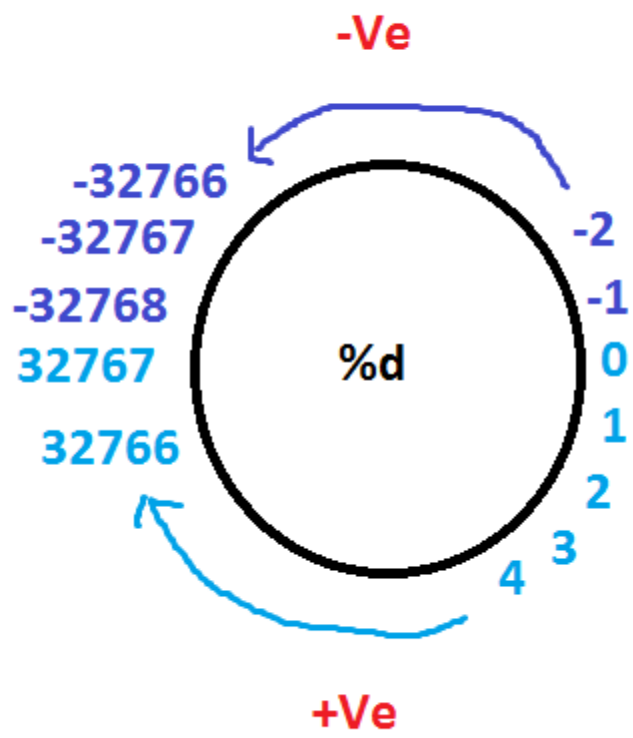
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
2:20 PM
29-Nov-24
```

```
TC
a=-32768
a=-32766
a=4
a=32766
```

Int cycle:

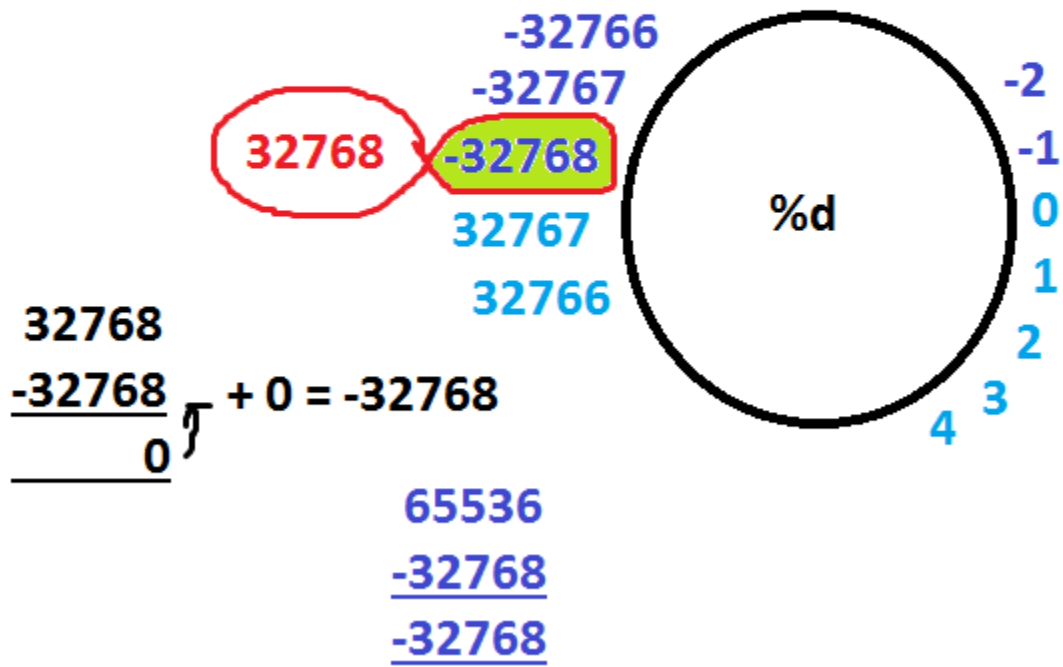
C & C++ working under 16 bit compilers $\Rightarrow 2^{16} \Rightarrow 65536$ / signed int -32768 to +32767
unsigned int 0 to 65535

signed int cycle



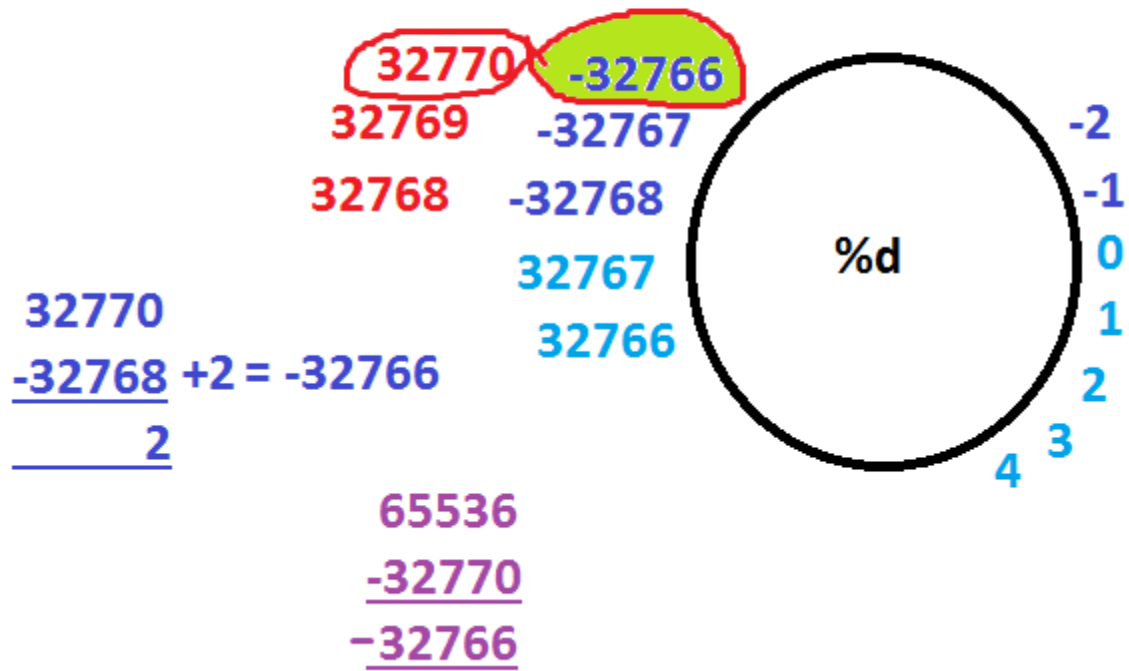
signed int cycle

int a=32768;



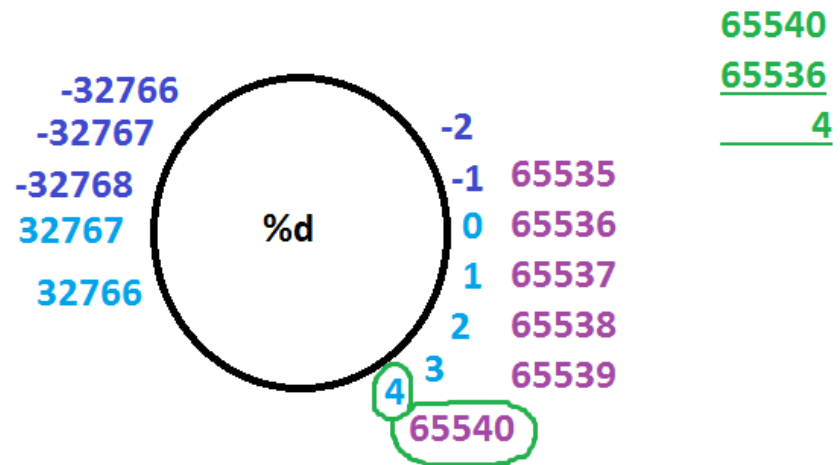
signed int cycle

int a = 32770;



signed int cycle

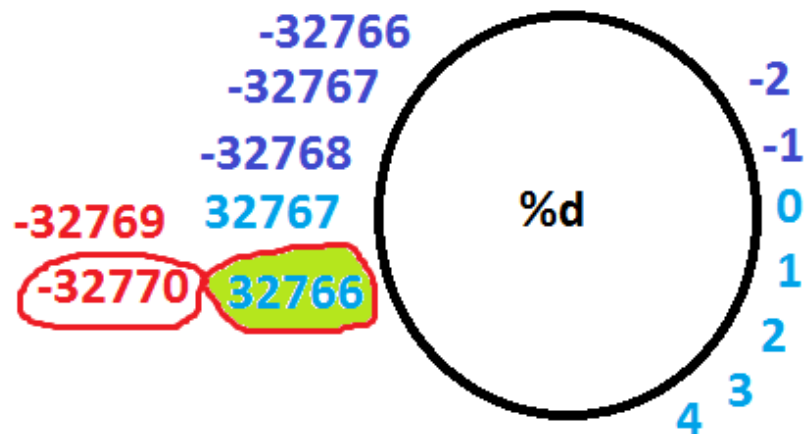
int a = 65540;

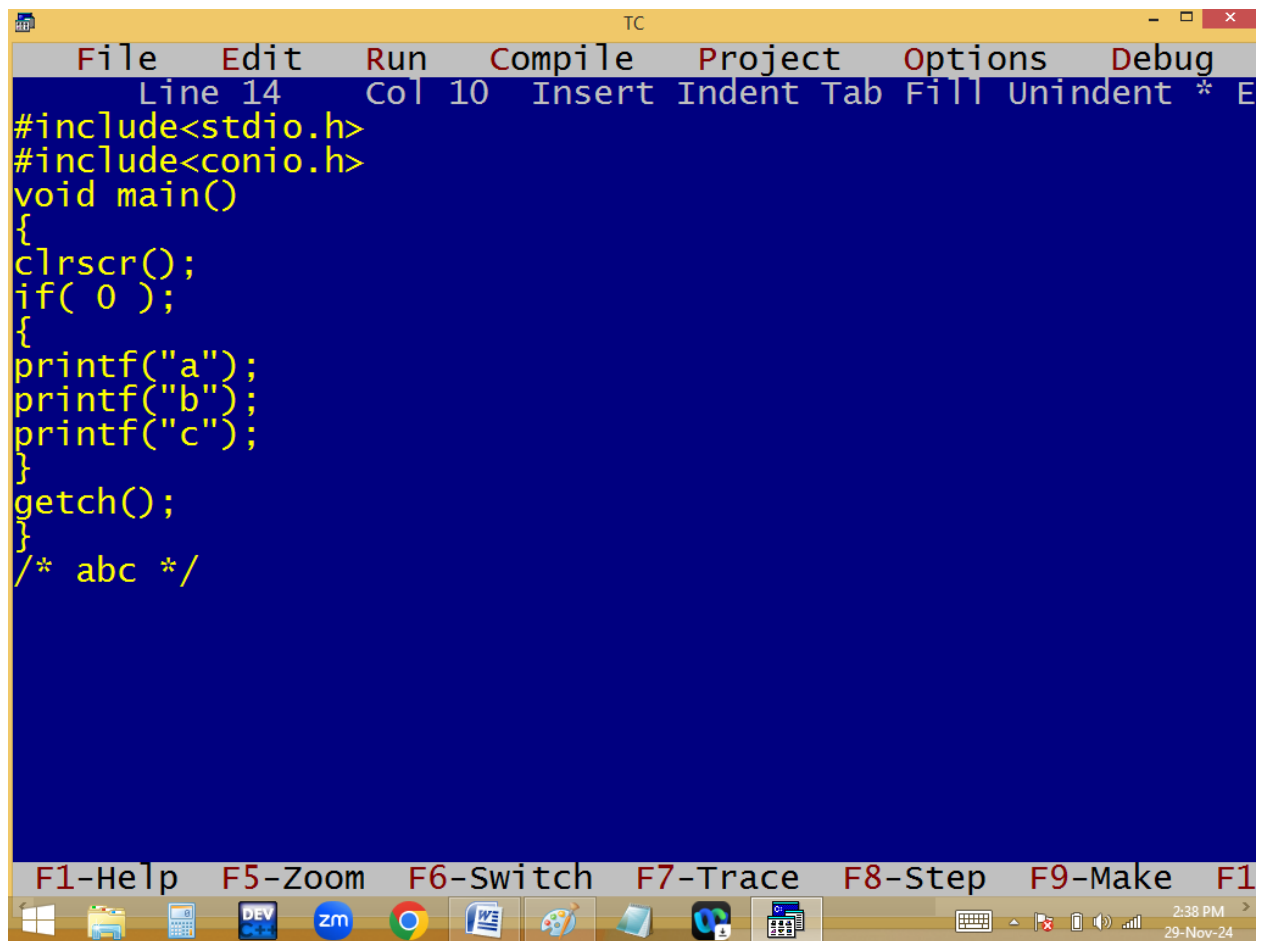


signed int cycle

int a = -32770;

65536
-32770

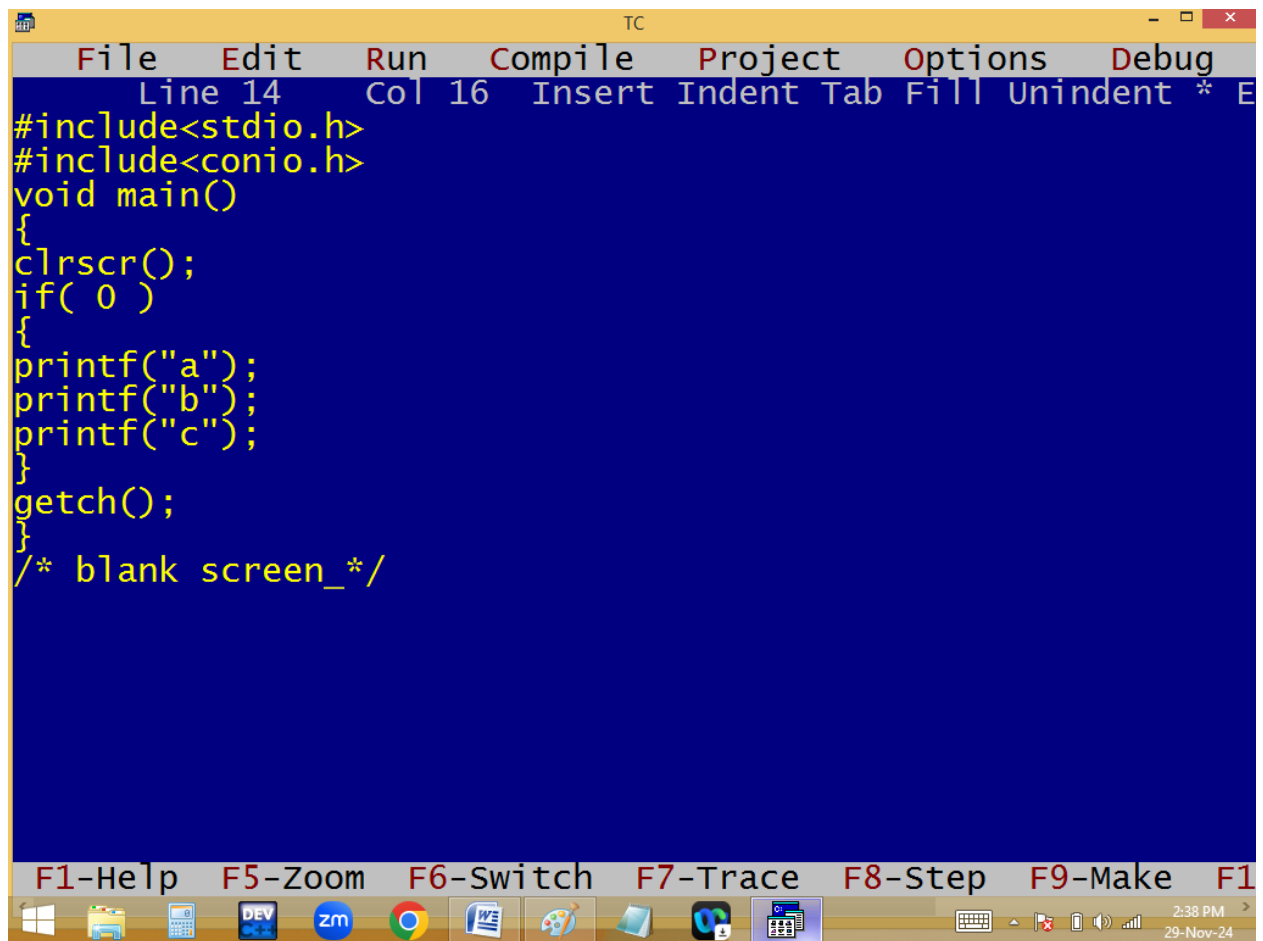




The image shows a screenshot of the Turbo C++ (TC) integrated development environment. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar indicates "Line 14", "Col 10", and lists editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and a toggle for "E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( 0 );
{
printf("a");
printf("b");
printf("c");
}
getch();
}
/* abc */
```

At the bottom of the window, there is a toolbar with icons for various functions. Above the toolbar, a row of function key shortcuts is displayed: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Exit". The Windows taskbar is visible at the very bottom, showing the Start button, several application icons (including File Explorer, Calculator, DEV C++, Zoom, Chrome, Word, and Paint), and the system tray with the date and time "2:38 PM 29-Nov-24".



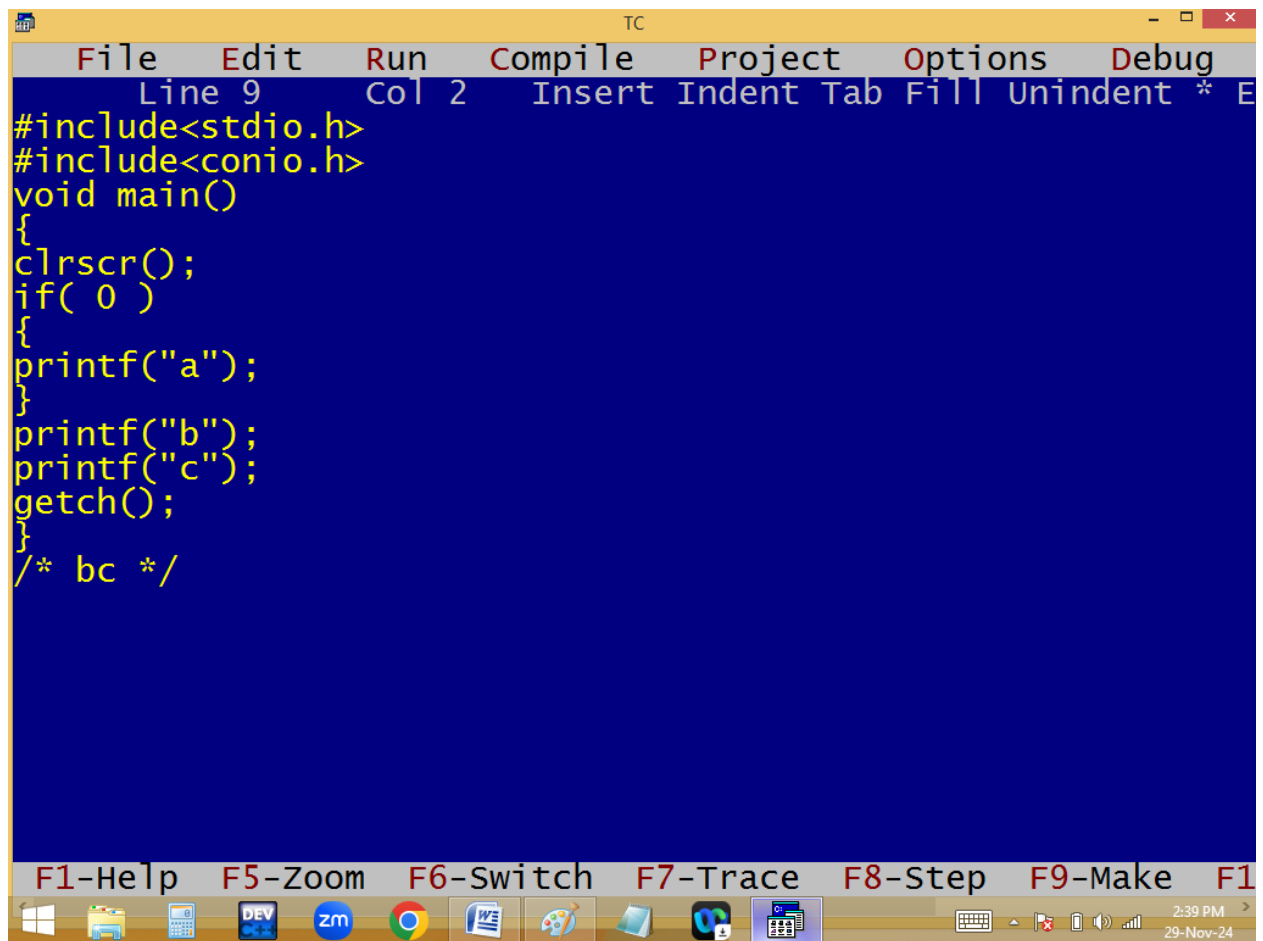
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 14", "Col 16", and lists editing modes: "Insert", "Indent", "Tab", "Fill", "Unindent", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( 0 )
{
printf("a");
printf("b");
printf("c");
}
getch();
}
/* blank screen_*/
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and the Turbo C++ application. The system clock in the bottom right corner shows "2:38 PM" and "29-Nov-24".

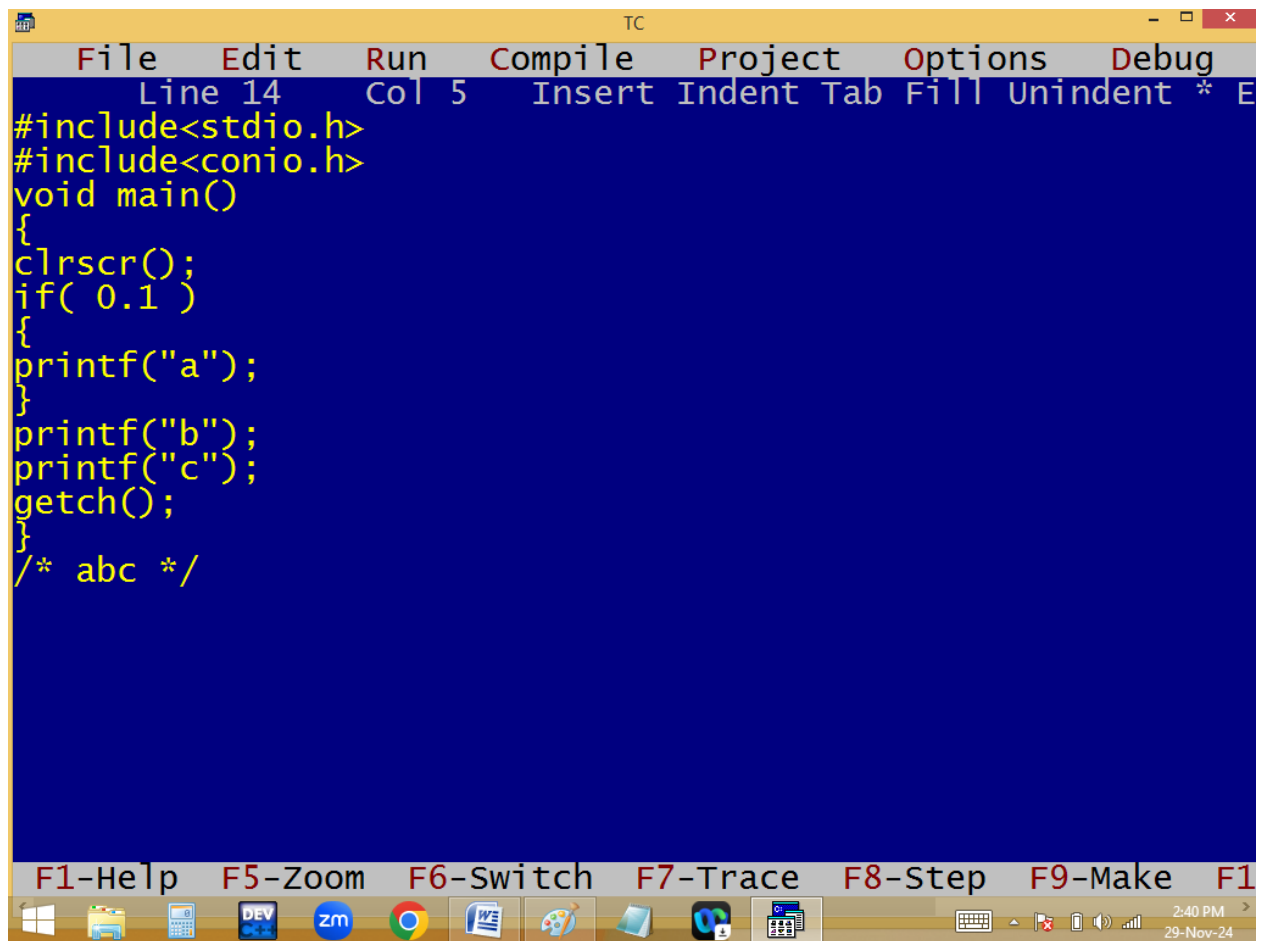

```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 6 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( 0 )
printf("a");
printf("b");
printf("c");
getch();
}
/* bc_*/

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
2:39 PM
29-Nov-24
```



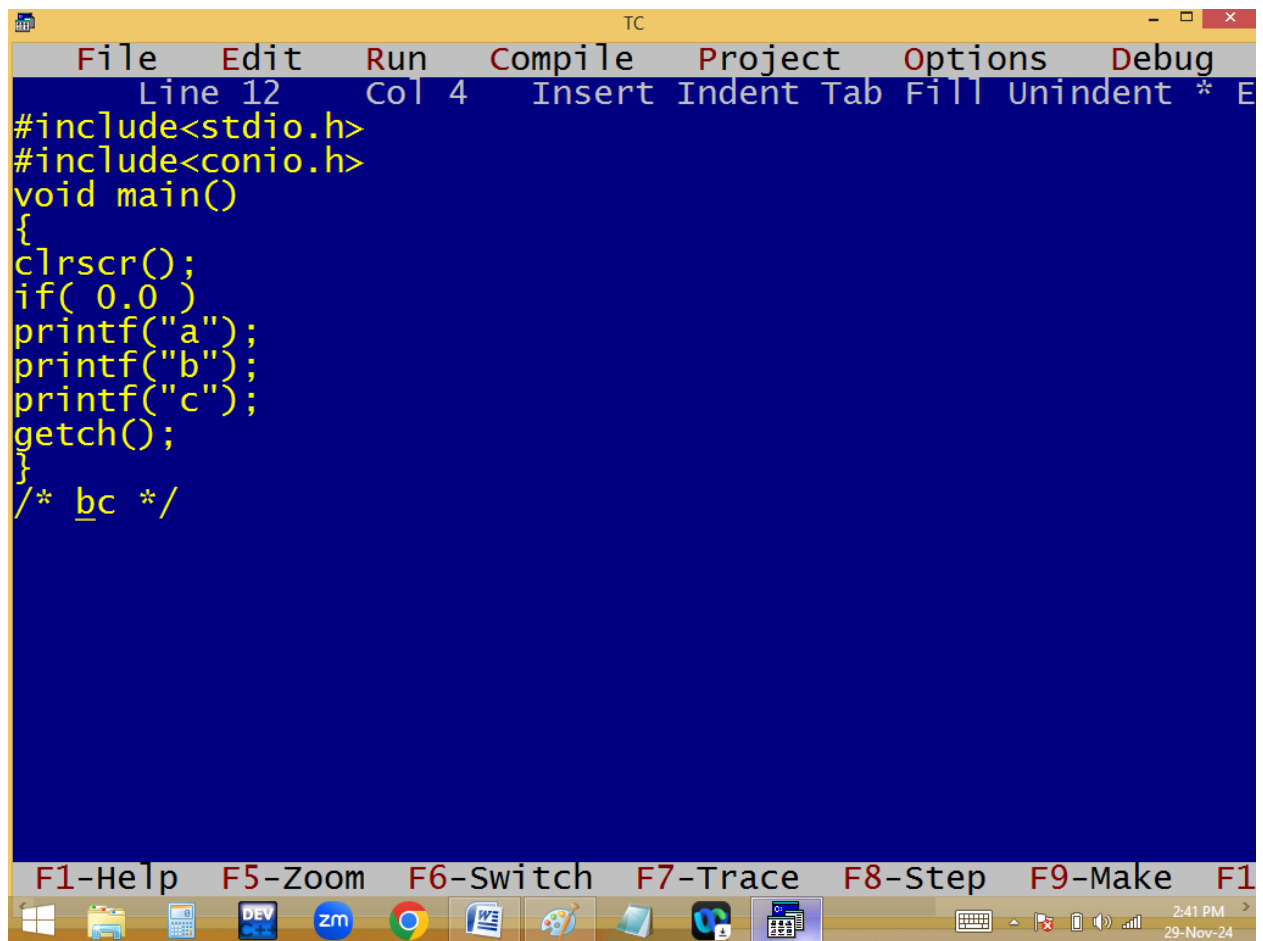
```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 2 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( 0 )
{
printf("a");
}
printf("b");
printf("c");
getch();
}
/* bc */

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
2:39 PM 29-Nov-24
```



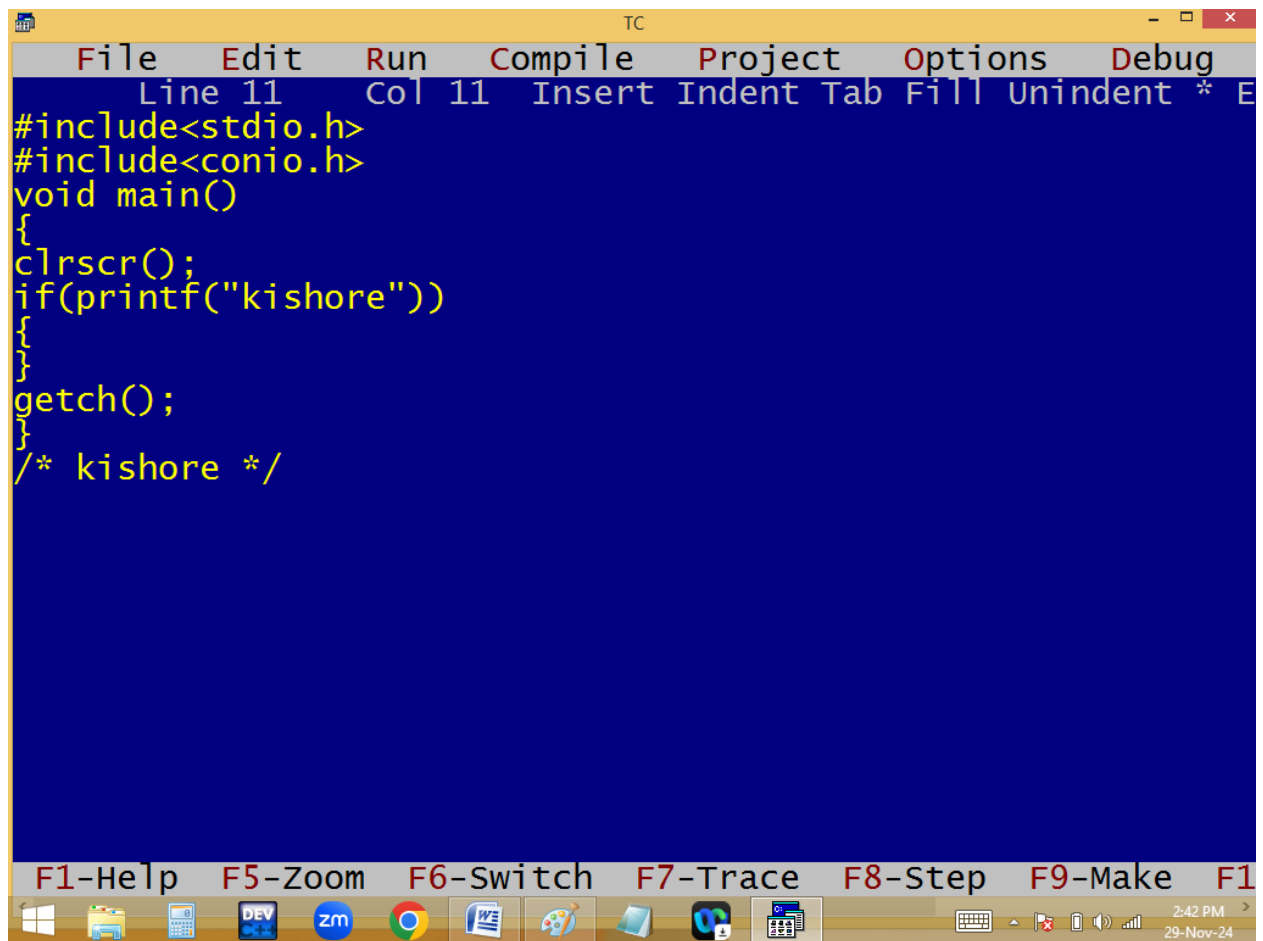
```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 5 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( 0.1 )
{
printf("a");
}
printf("b");
printf("c");
getch();
}
/* abc */

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
2:40 PM
29-Nov-24
```



```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 4 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( 0.0 )
printf("a");
printf("b");
printf("c");
getch();
}
/* bc */

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
2:41 PM
29-Nov-24
```



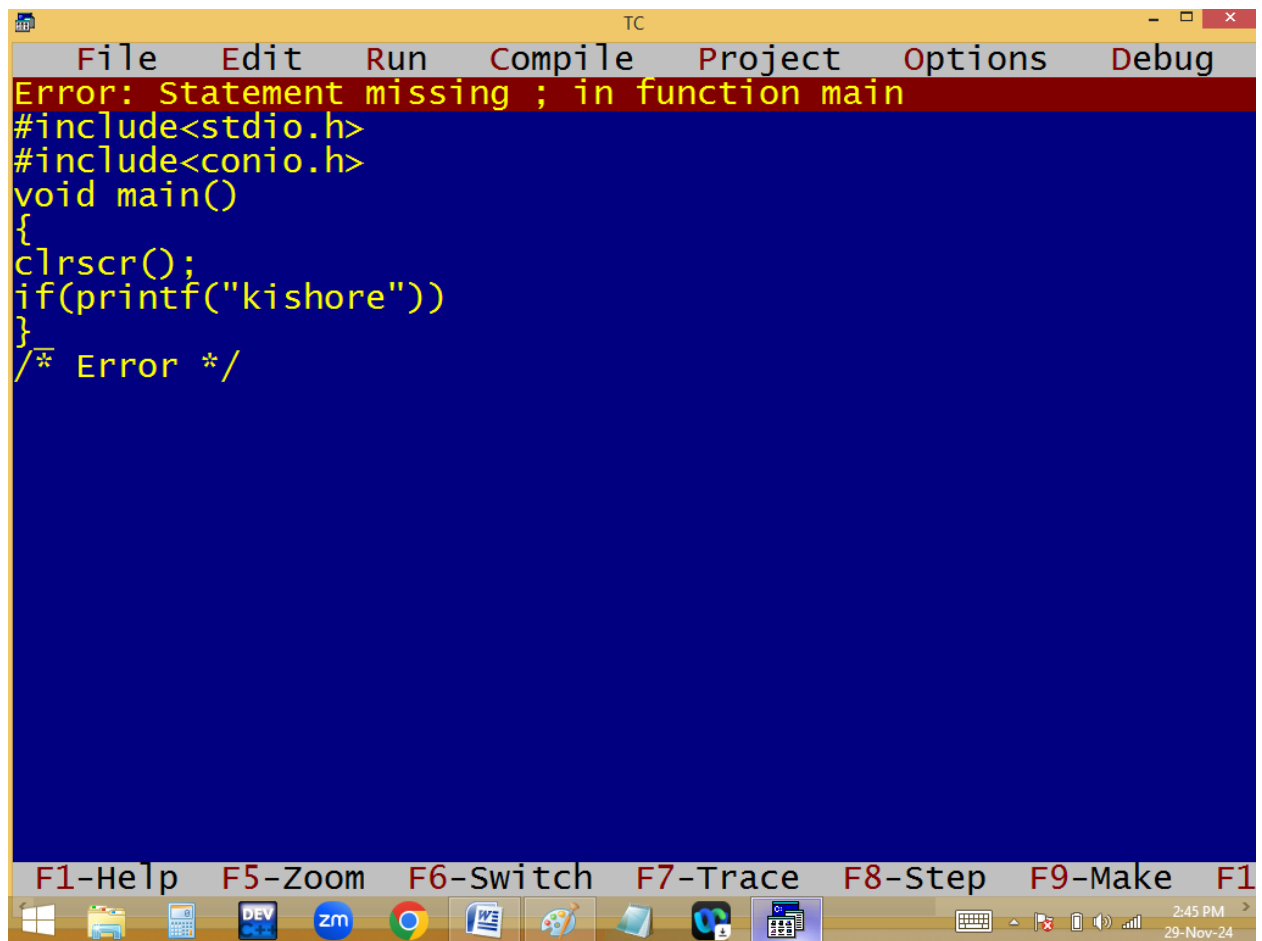
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 11", "Col 11", and various editing options like "Insert", "Indent", "Tab", "Fill", "Unindent", and "E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(printf("kishore"))
{
}
getch();
}
/* kishore */
```

At the bottom of the window, there is a toolbar with icons for various functions and a row of function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other applications. The system clock in the bottom right corner indicates the time is 2:42 PM on 29-Nov-24.

```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 12 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(printf("kishore"))
getch();
}
/* Kishore */

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
2:44 PM 29-Nov-24
```

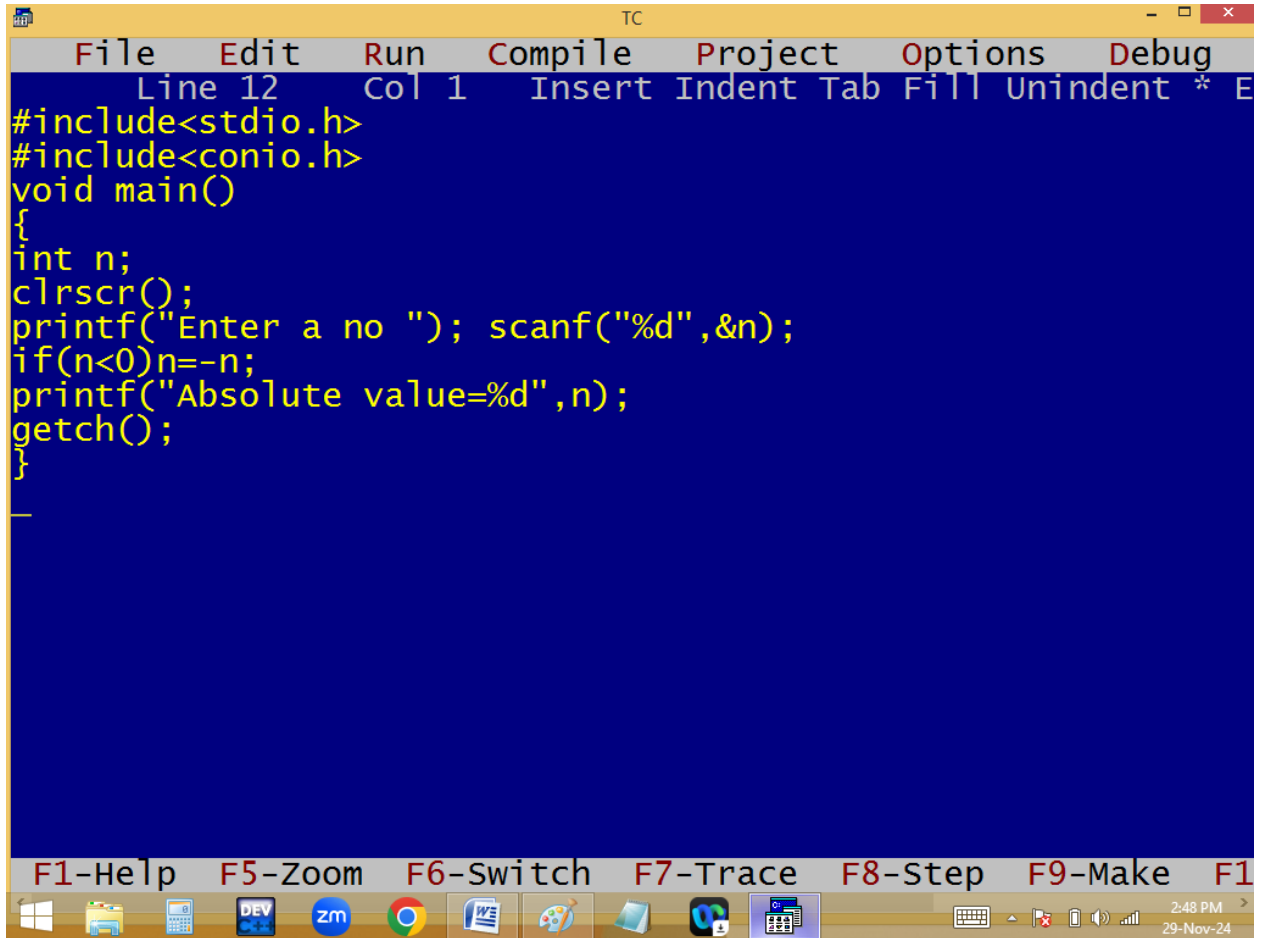


The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code editor states: "Error: Statement missing ; in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(printf("kishore"))
}
/* Error */
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows various application icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder icon. The system tray on the right indicates the time is 2:45 PM on 29-Nov-24.

Finding absolute value [always +Ve]:

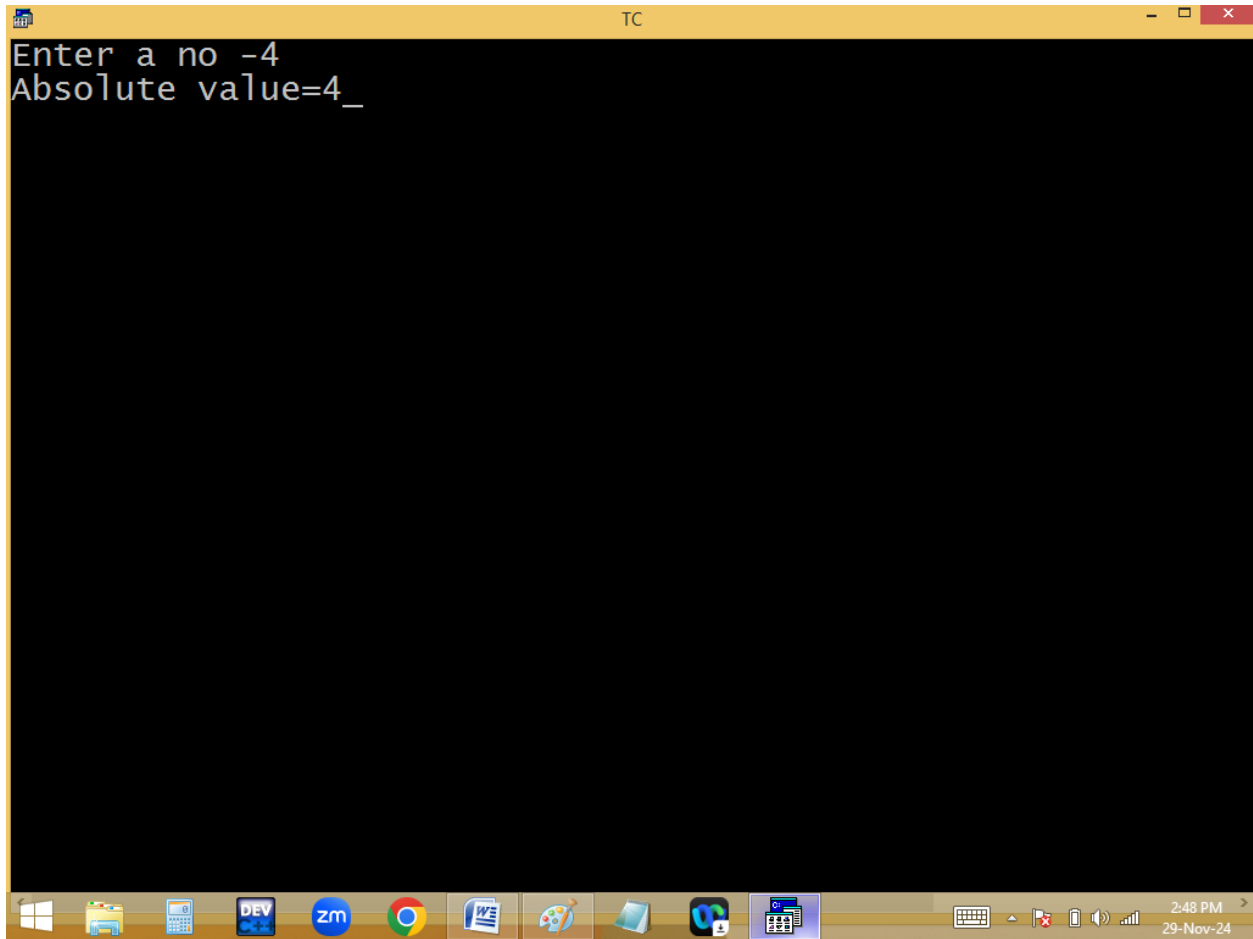


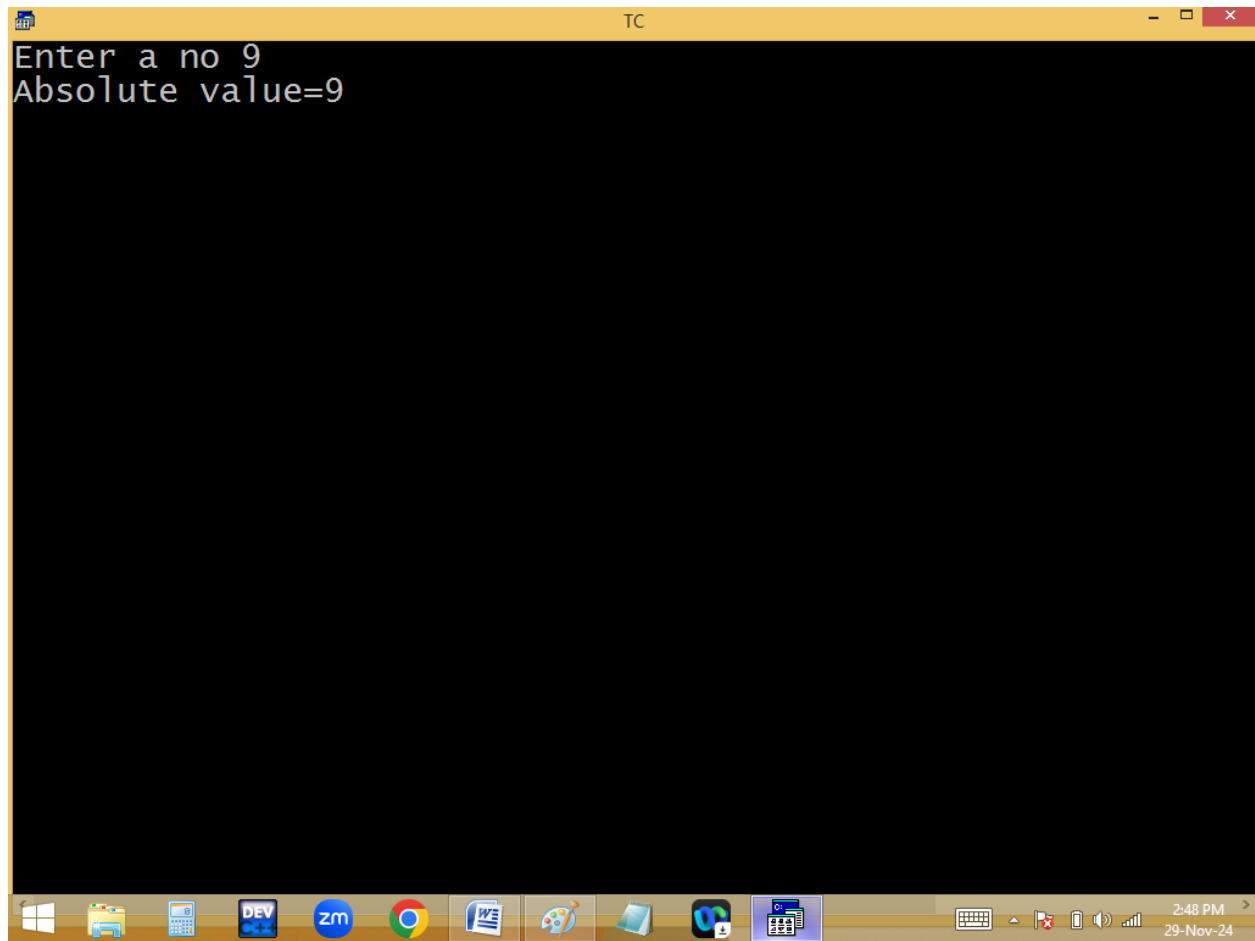
The screenshot shows the Turbo C++ (TC) IDE interface. The main window displays a C program designed to calculate the absolute value of an integer. The code includes standard headers for input/output and console operations. It defines a `main` function where an integer `n` is declared, the screen is cleared, and the user is prompted to enter a number. The program then checks if the number is negative; if so, it negates it. Finally, it prints the absolute value and waits for a key press before exiting.

```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
if(n<0)n=-n;
printf("Absolute value=%d",n);
getch();
}
```

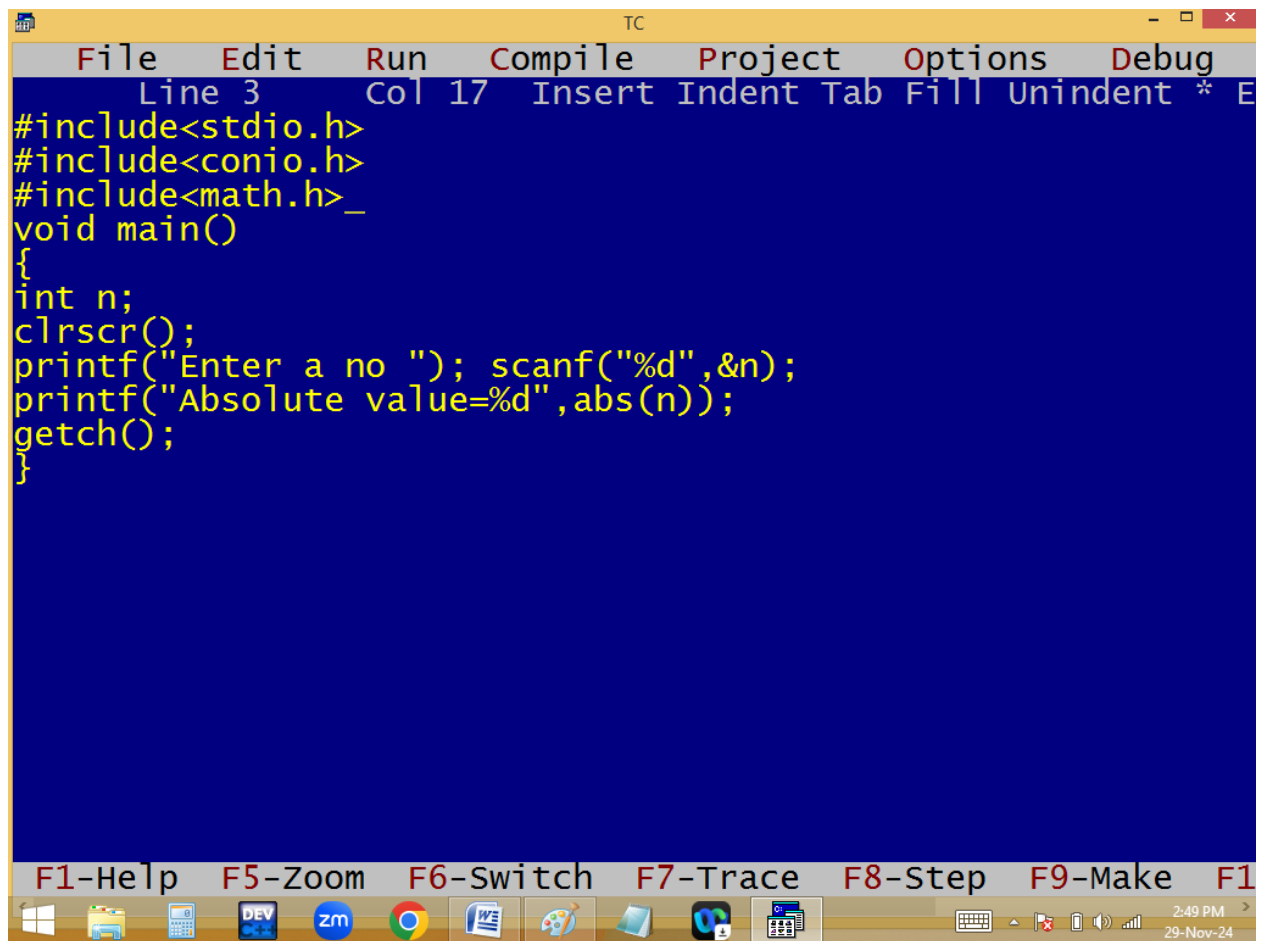
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

2:48 PM
29-Nov-24





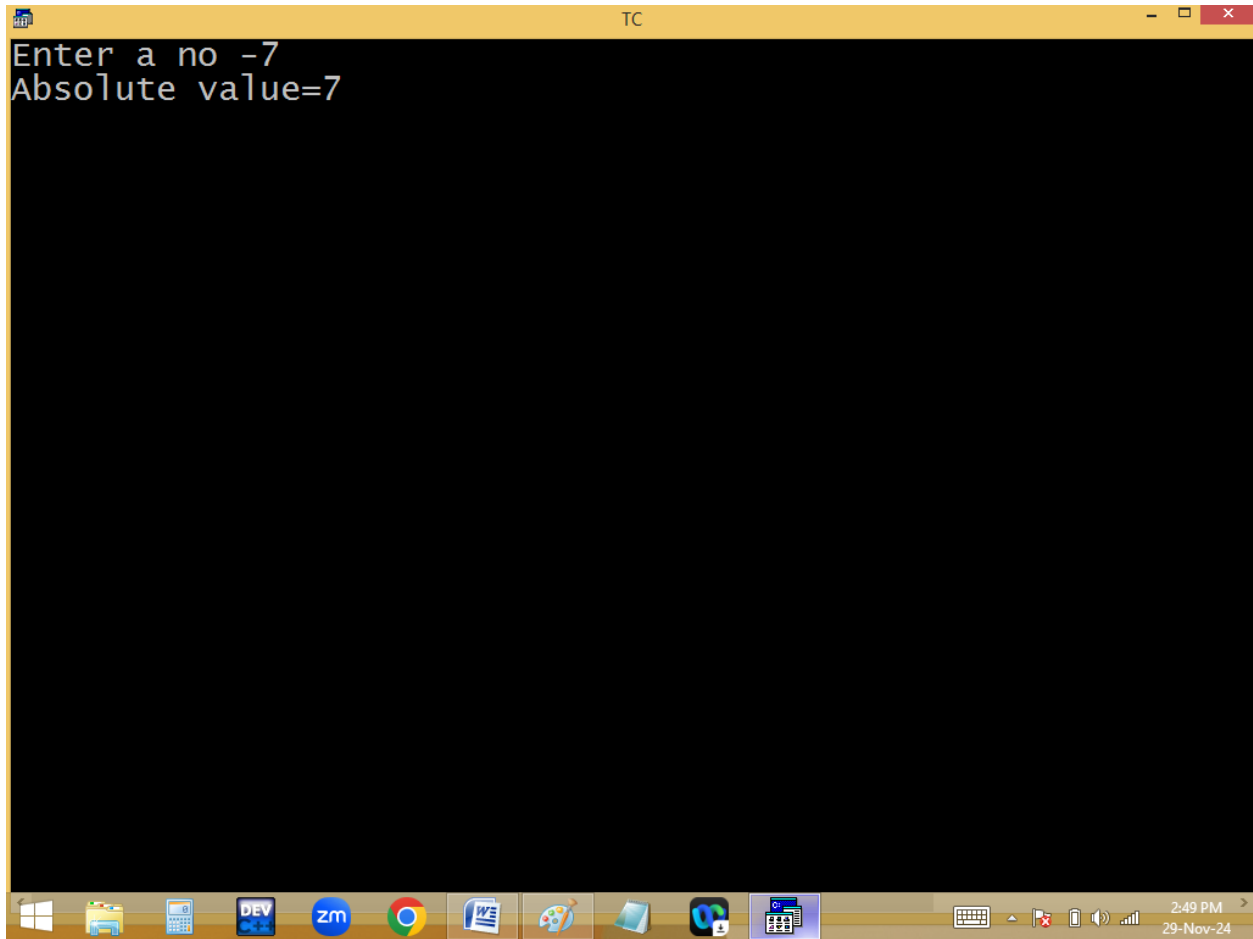
Using abs():



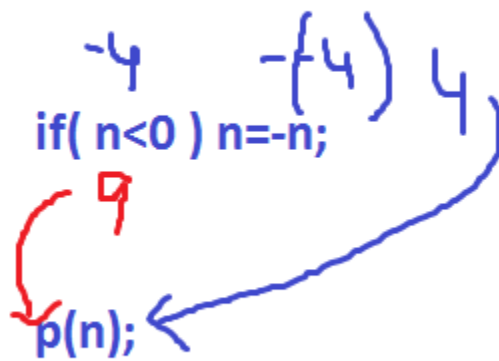
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 3", "Col 17", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>_
void main()
{
int n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
printf("Absolute value=%d",abs(n));
getch();
}
```

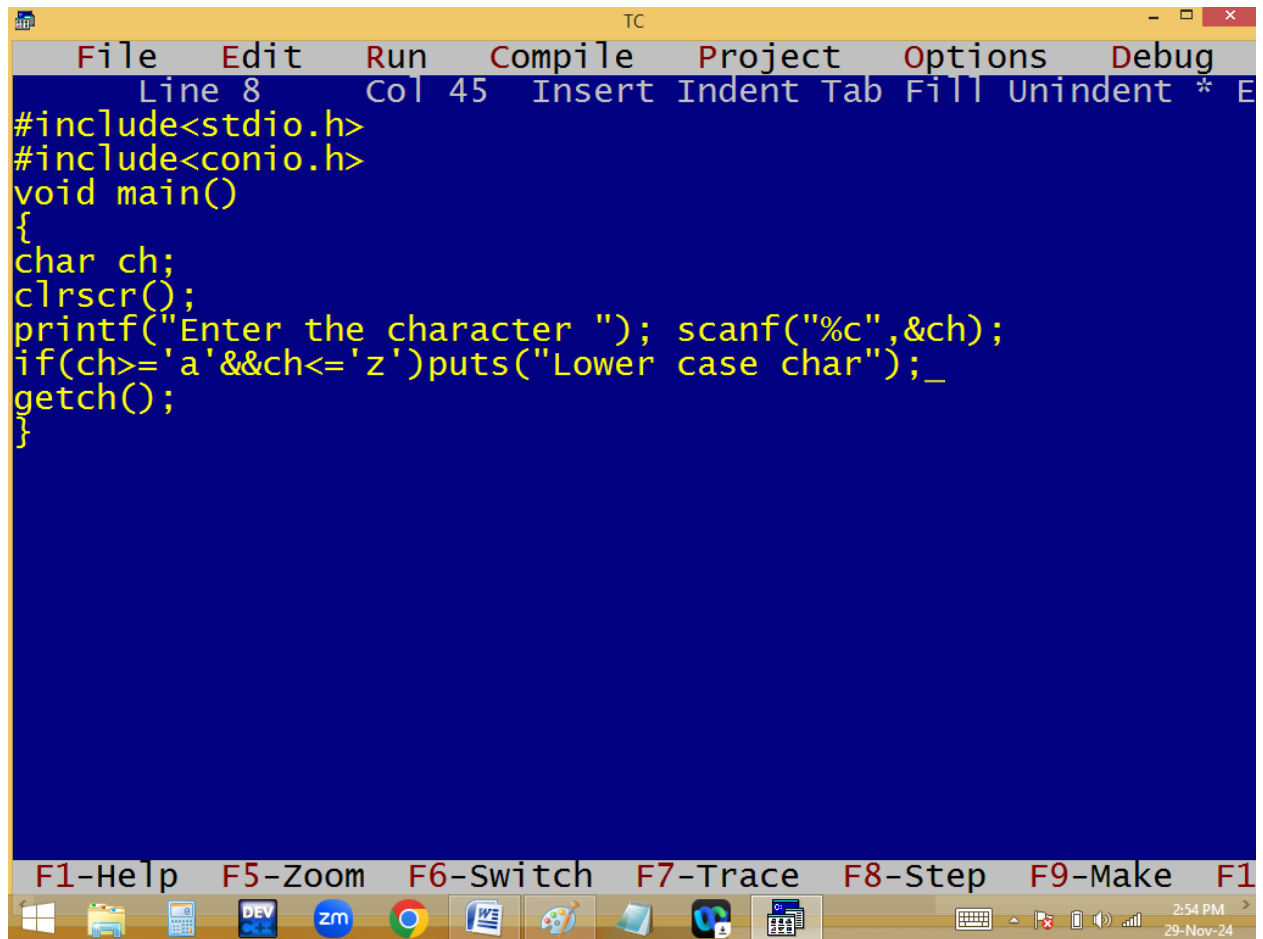
At the bottom of the window, there is a toolbar with icons for various functions. Below the toolbar, a row of function key shortcuts is displayed: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Exit". The Windows taskbar is visible at the very bottom, showing several application icons and the system clock indicating "2:49 PM" on "29-Nov-24".



```
TC
Enter a no 8
Absolute value=8
```



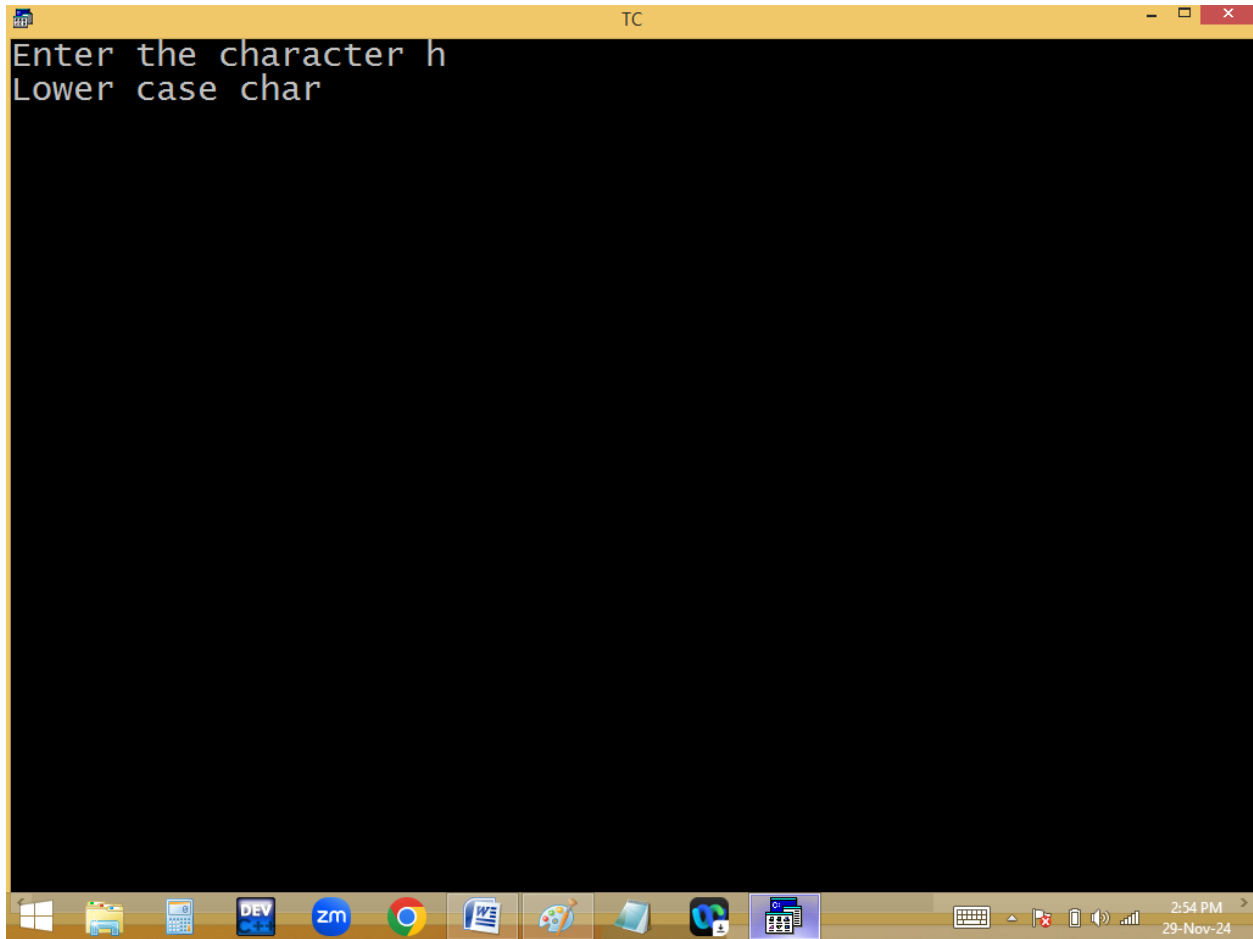
Finding lower case alphabet or not?

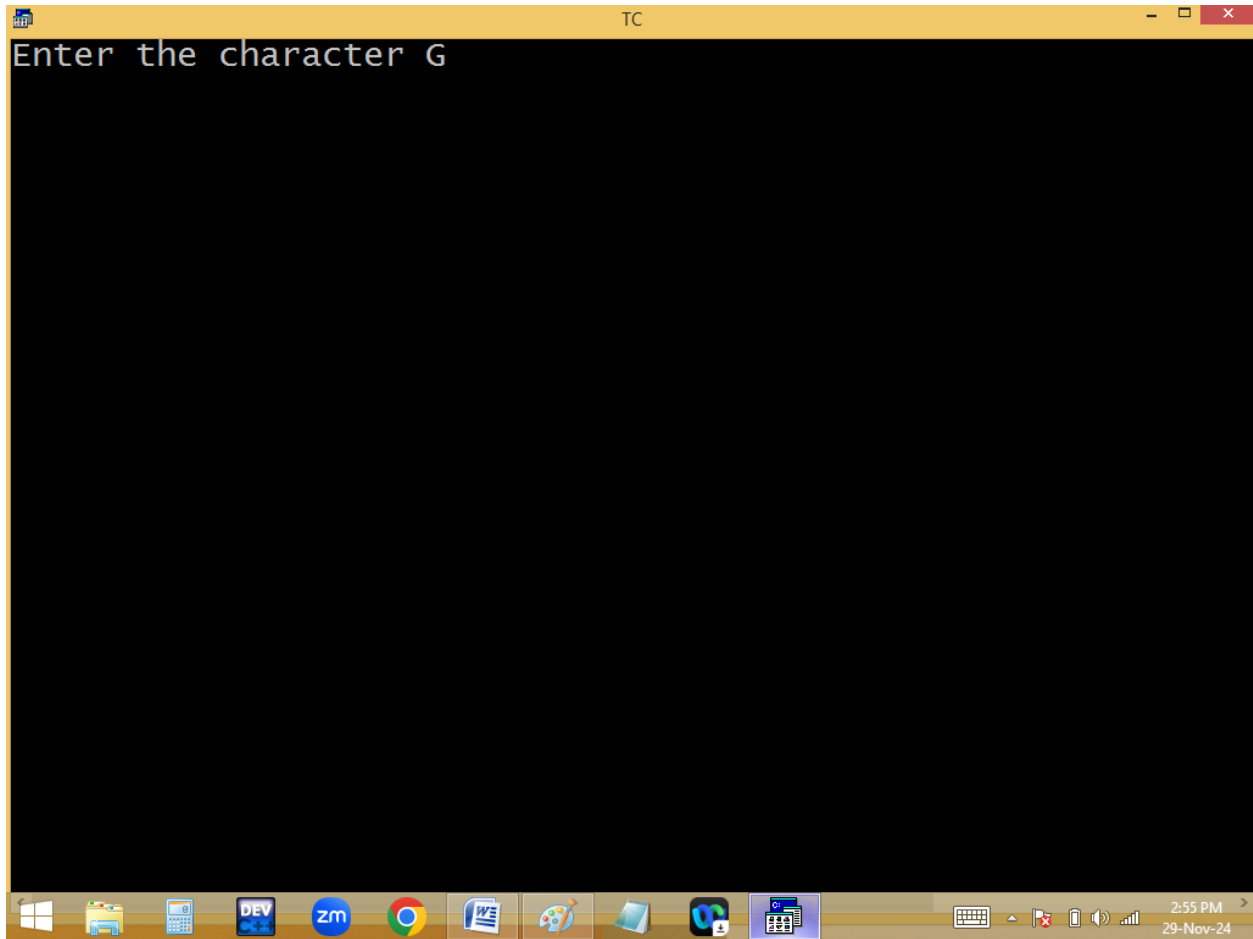


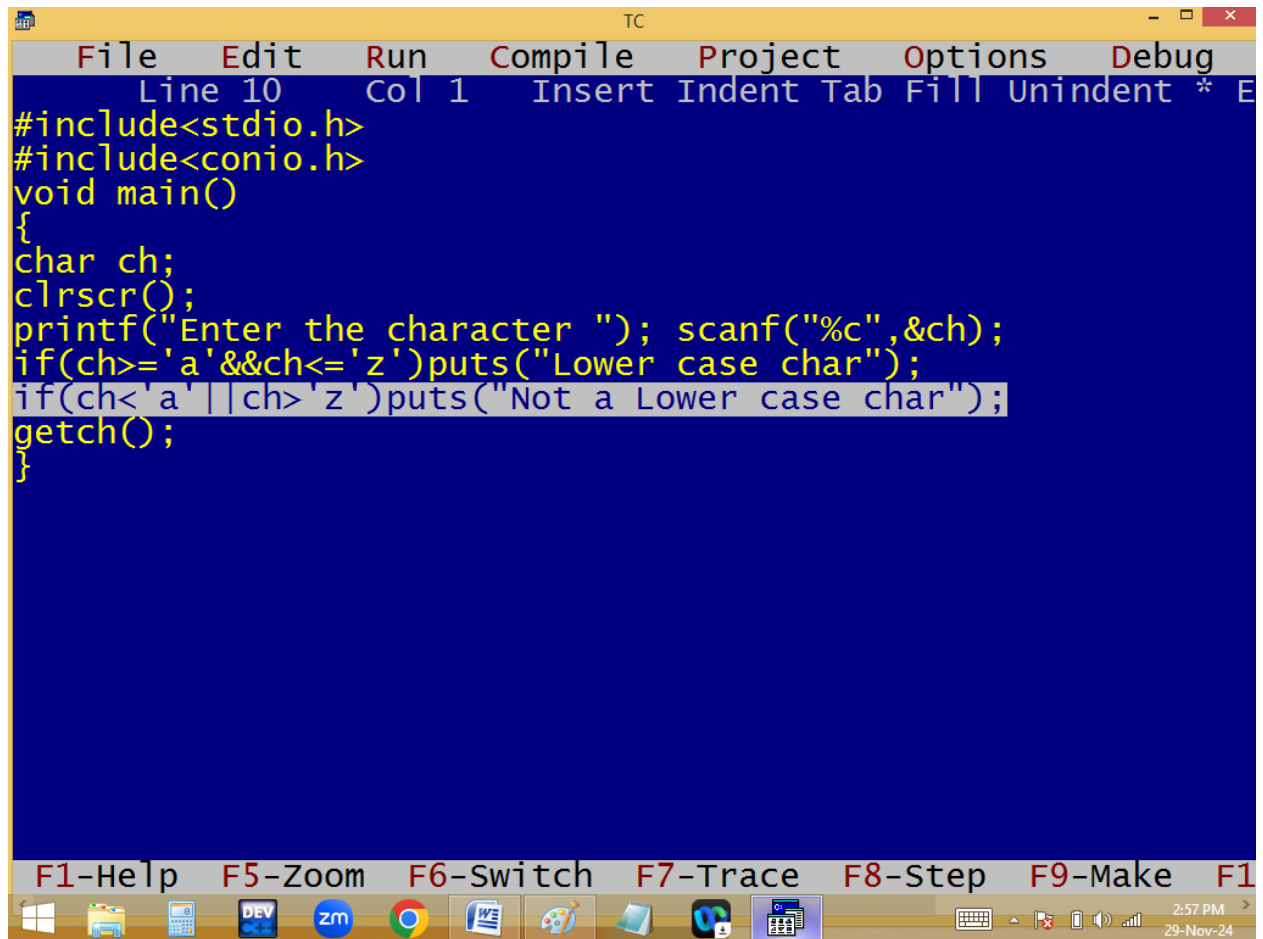
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 8", "Col 45", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a'&&ch<='z')puts("Lower case char");_
getch();
}
```

At the bottom of the IDE window, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the IDE window is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder icon. The system tray on the right shows the date and time as "2:54 PM" and "29-Nov-24".



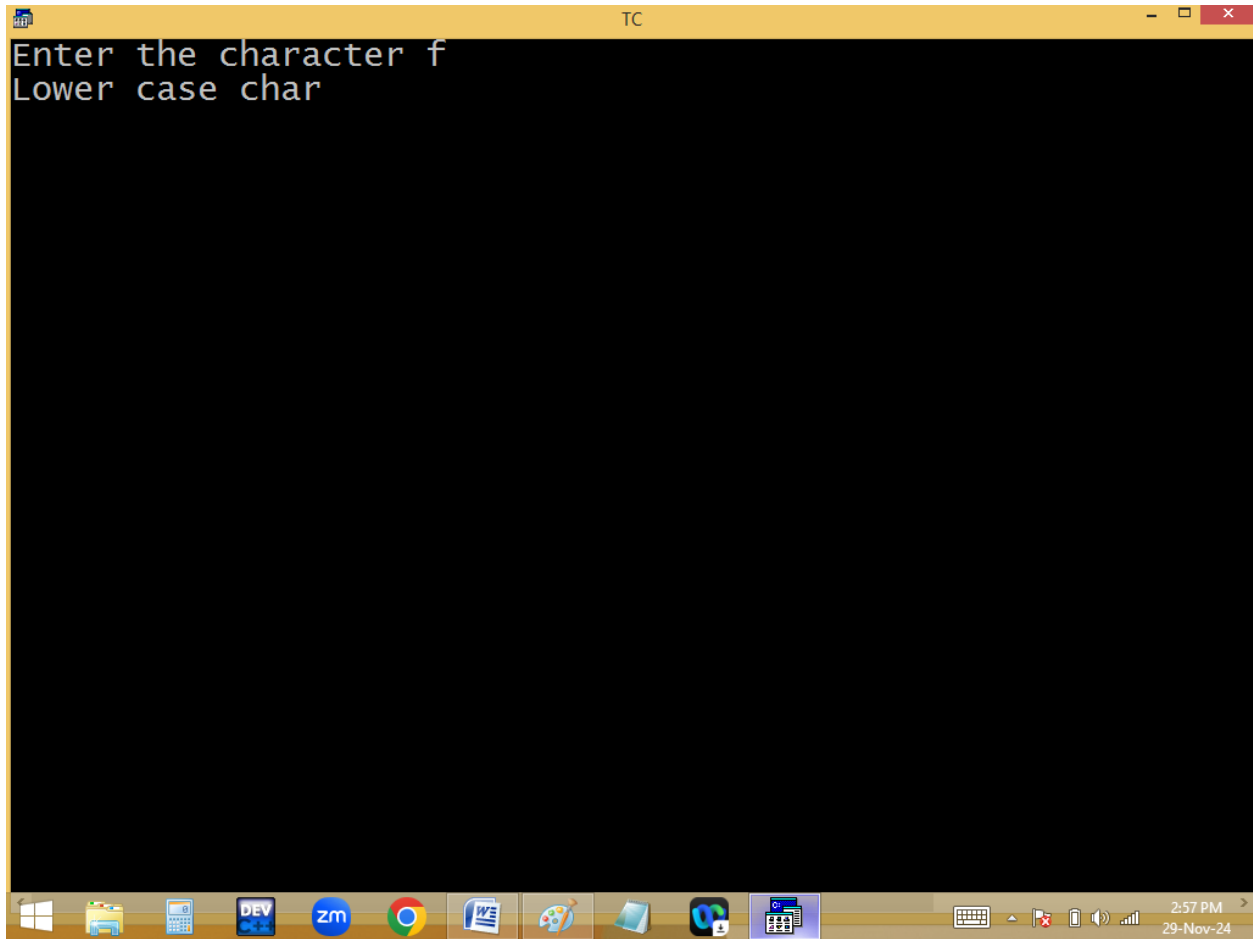


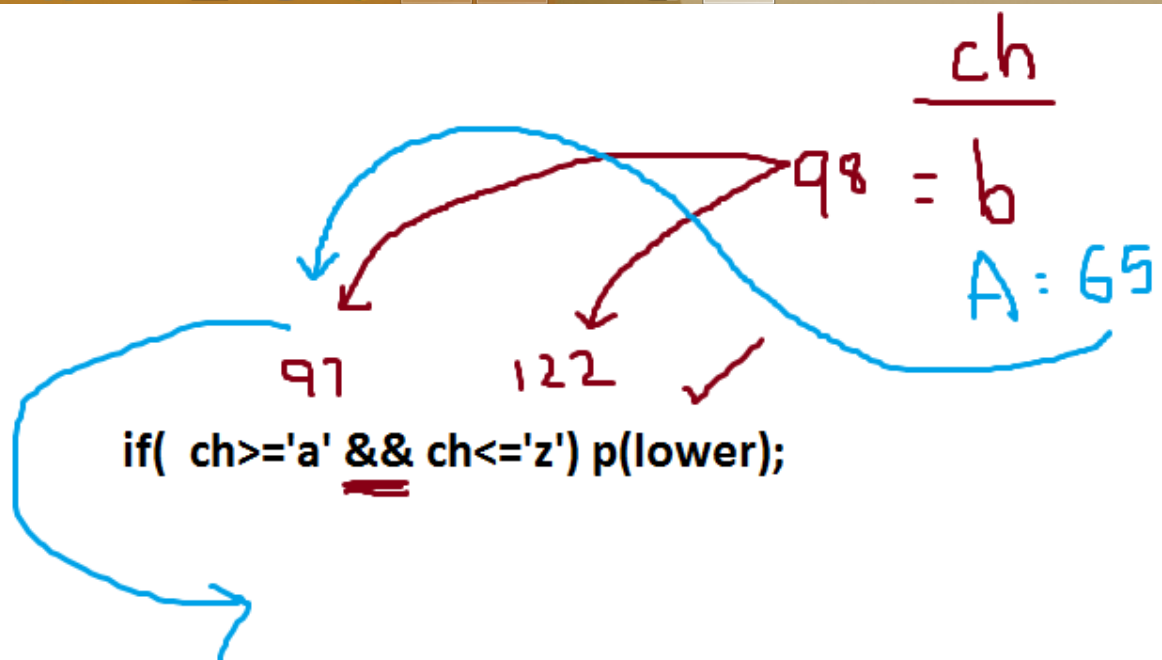
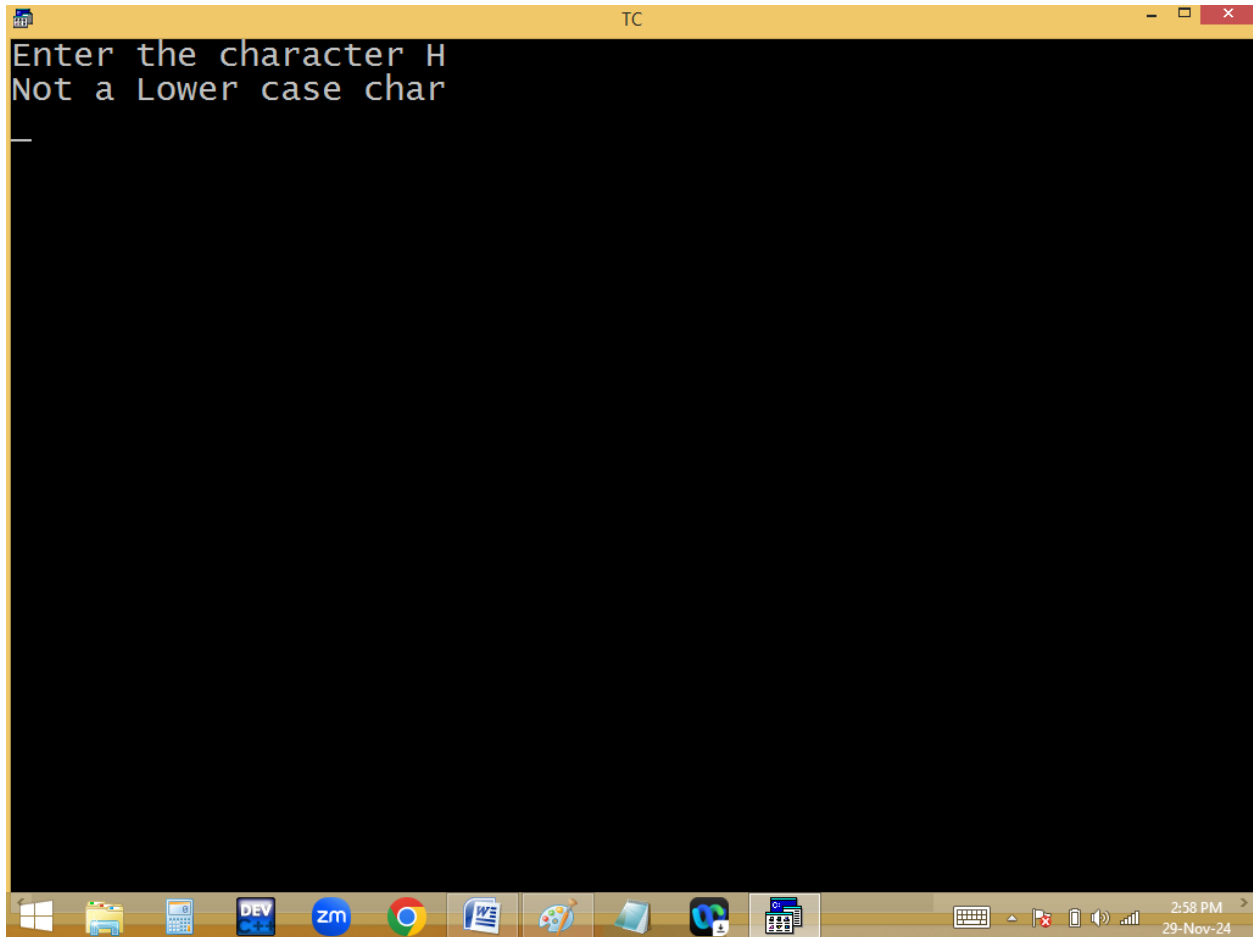


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 10", "Col 1", and "Insert Indent Tab Fill Unindent * E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a'&&ch<='z')puts("Lower case char");
if(ch<'a' || ch>'z')puts("Not a Lower case char");
getch();
}
```

The bottom status bar displays function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the very bottom shows various application icons and the system clock indicating "2:57 PM" on "29-Nov-24".





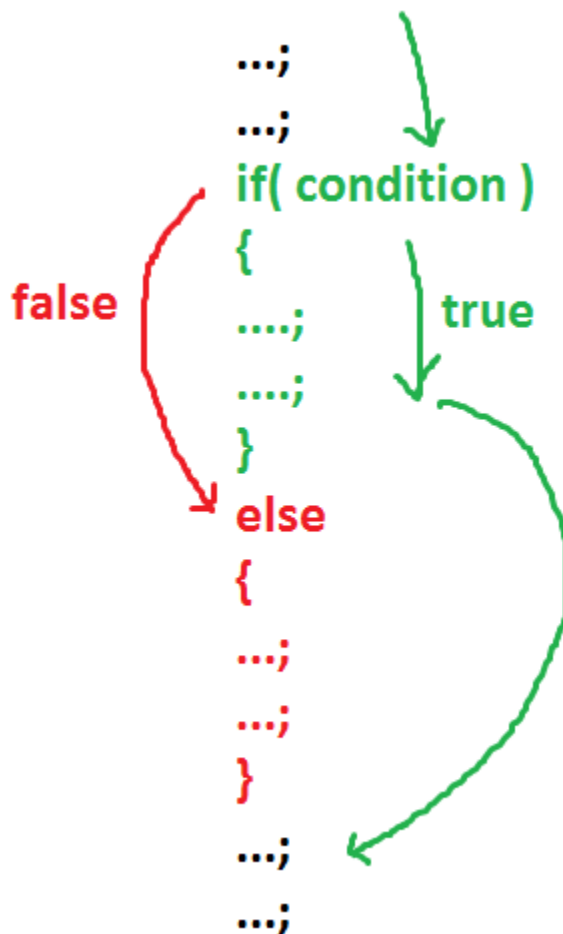
If..else:

When the program is having two options then go for if..else.

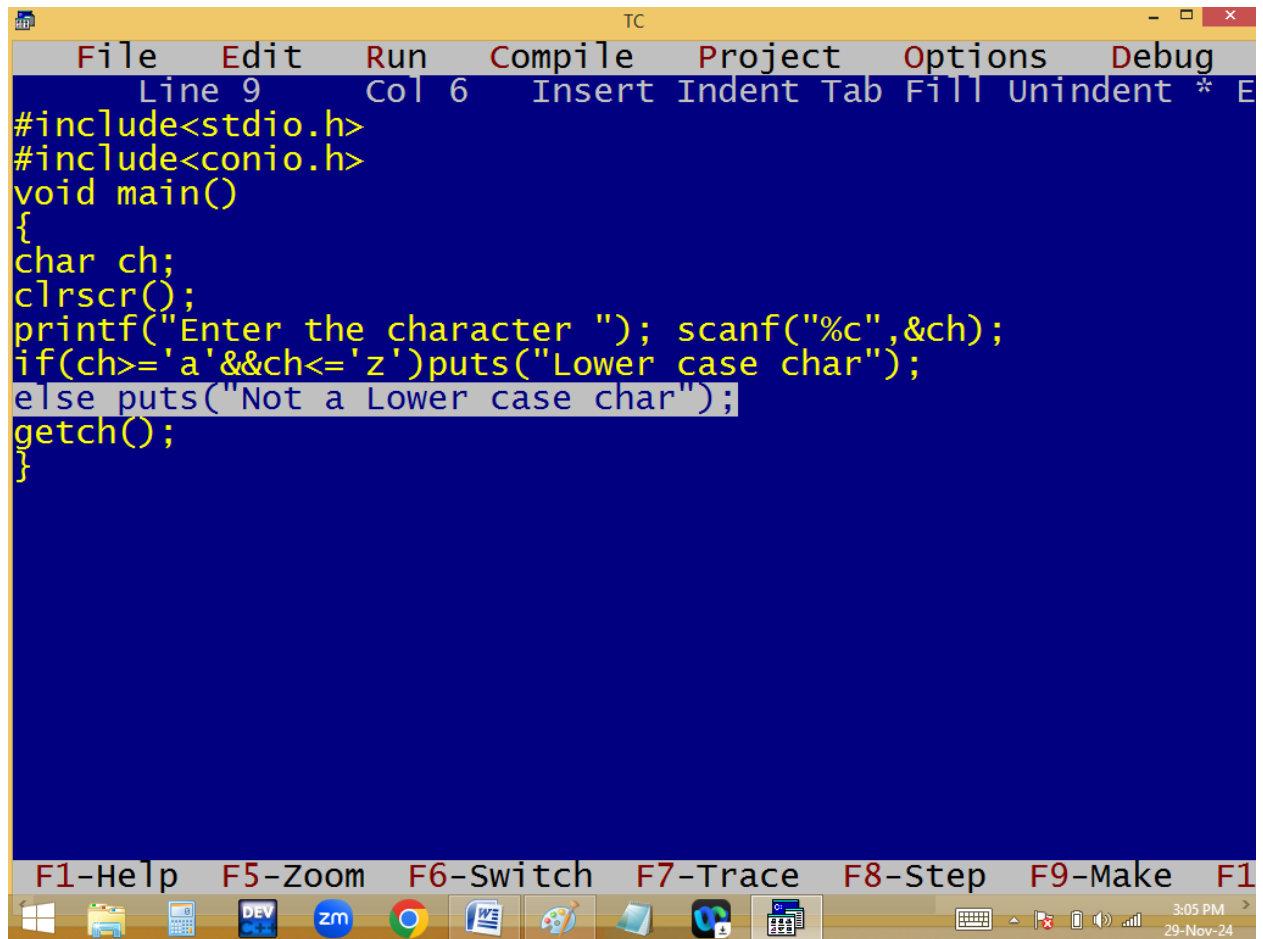
If condition true statements in if block { } are executed not going to check the else block.

If condition false statements in else block { } are working.

Here else doesn't require any condition.



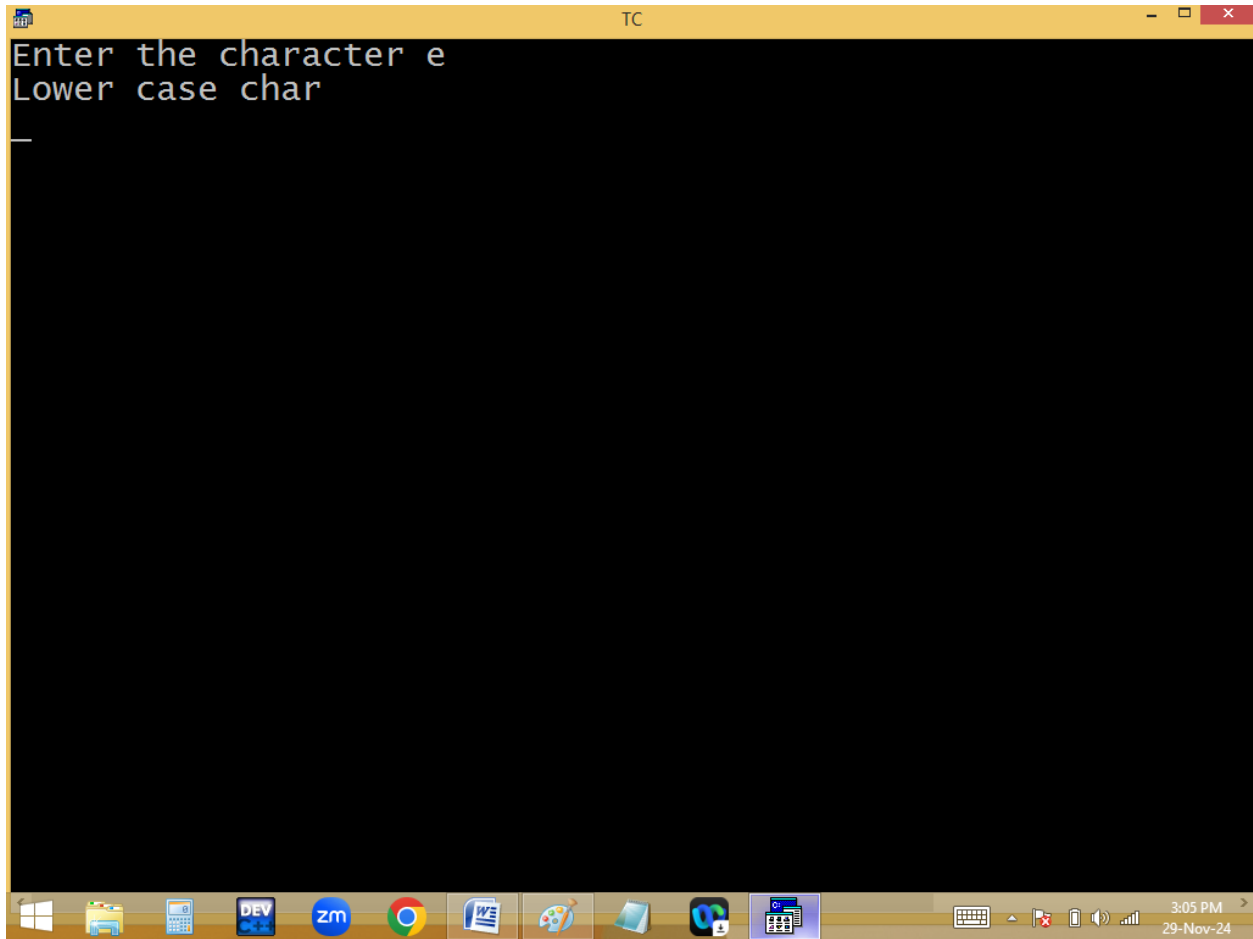
Finding lower case char or not using if...else:

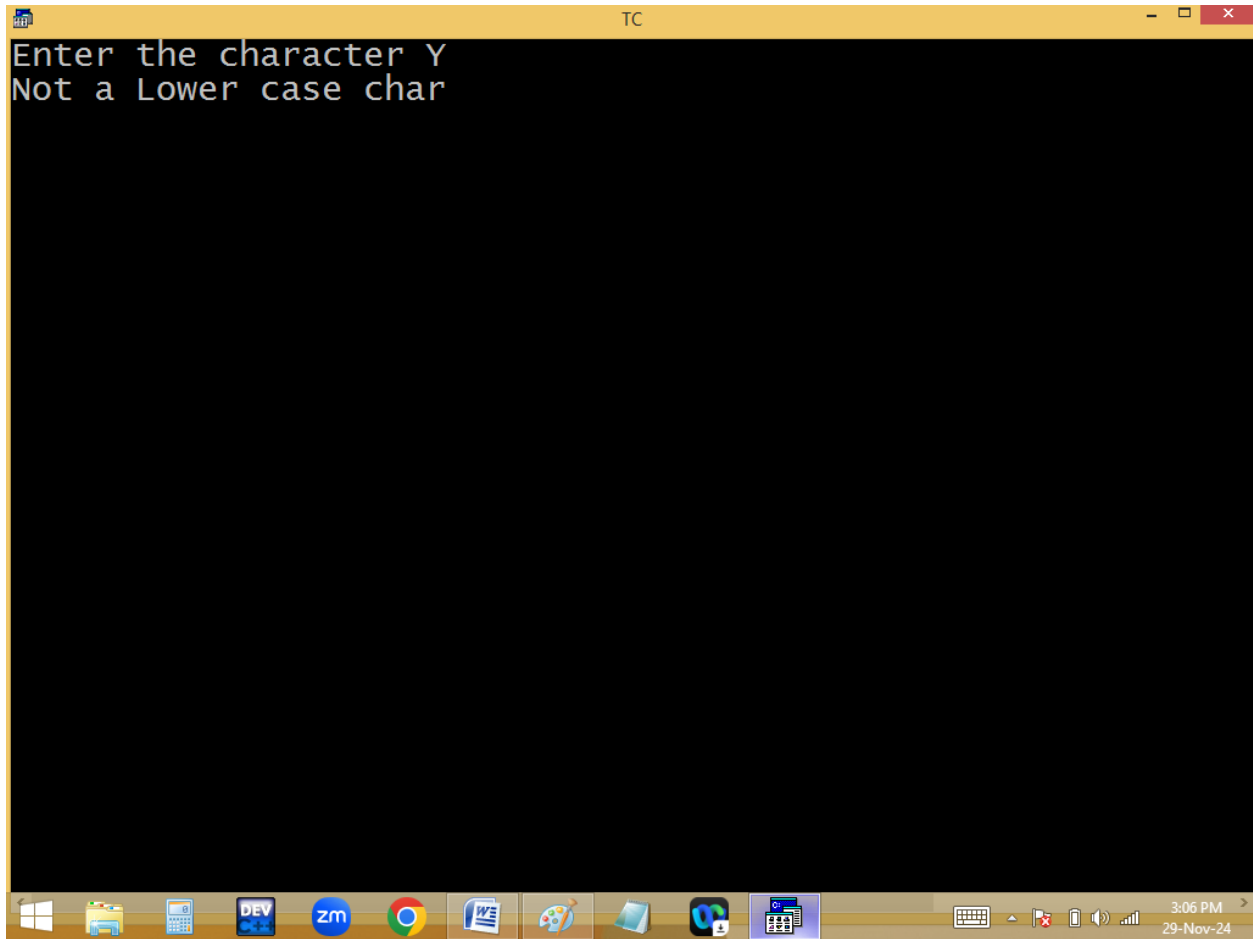


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following items: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Edit" menu, there is a submenu with the following options: Line 9, Col 6, Insert, Indent, Tab, Fill, Unindent, and * E. The main editing area has a dark blue background and contains the following C code:

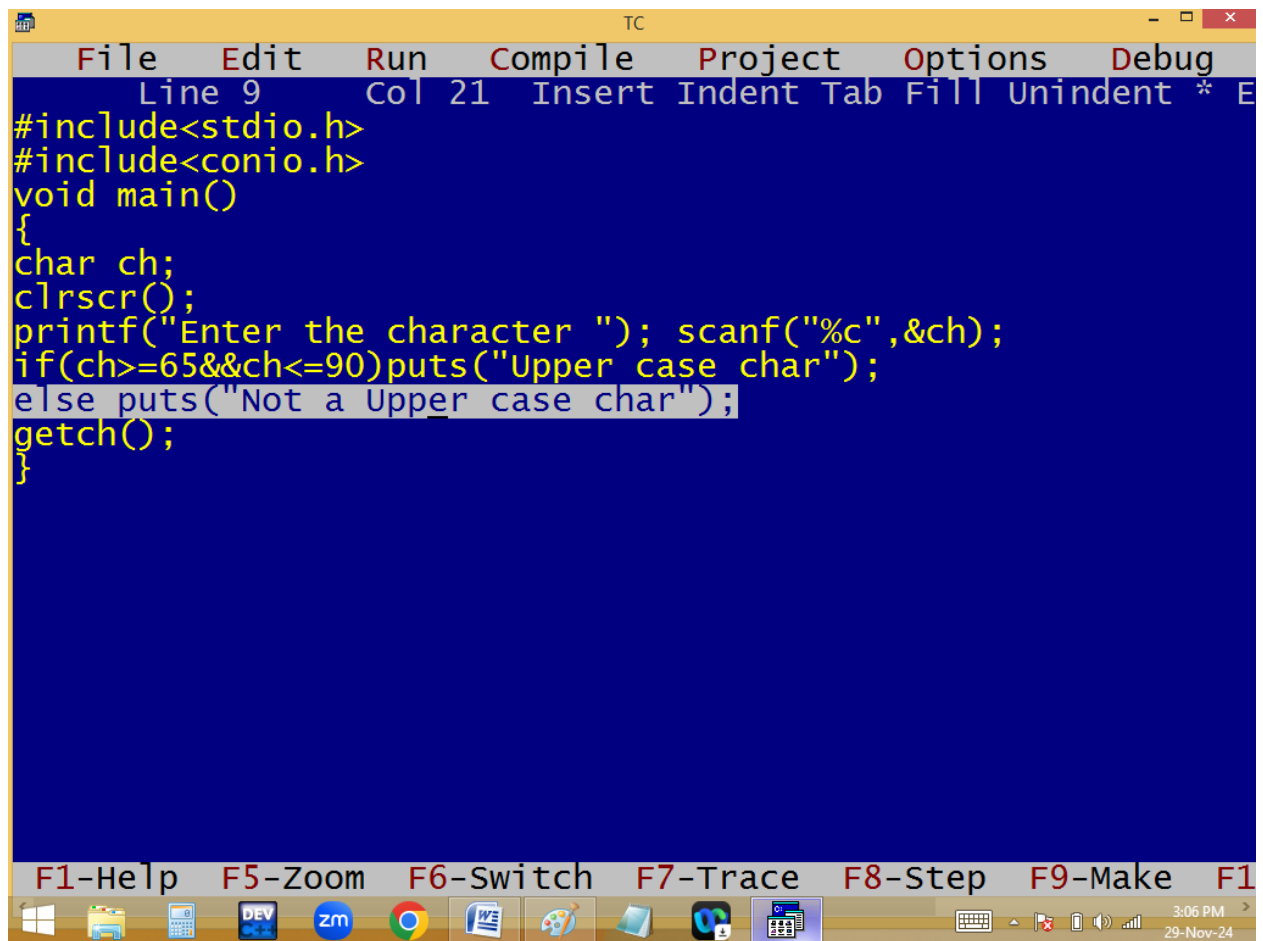
```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a'&&ch<='z')puts("Lower case char");
else puts("Not a Lower case char");
getch();
}
```

At the bottom of the window is a status bar with the following text: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F1. Below the status bar is a Windows taskbar with various icons, including the Start button, File Explorer, Calculator, DEV, zm, Google Chrome, Word, Paint, and a folder icon. The system clock in the bottom right corner shows 3:05 PM on 29-Nov-24.





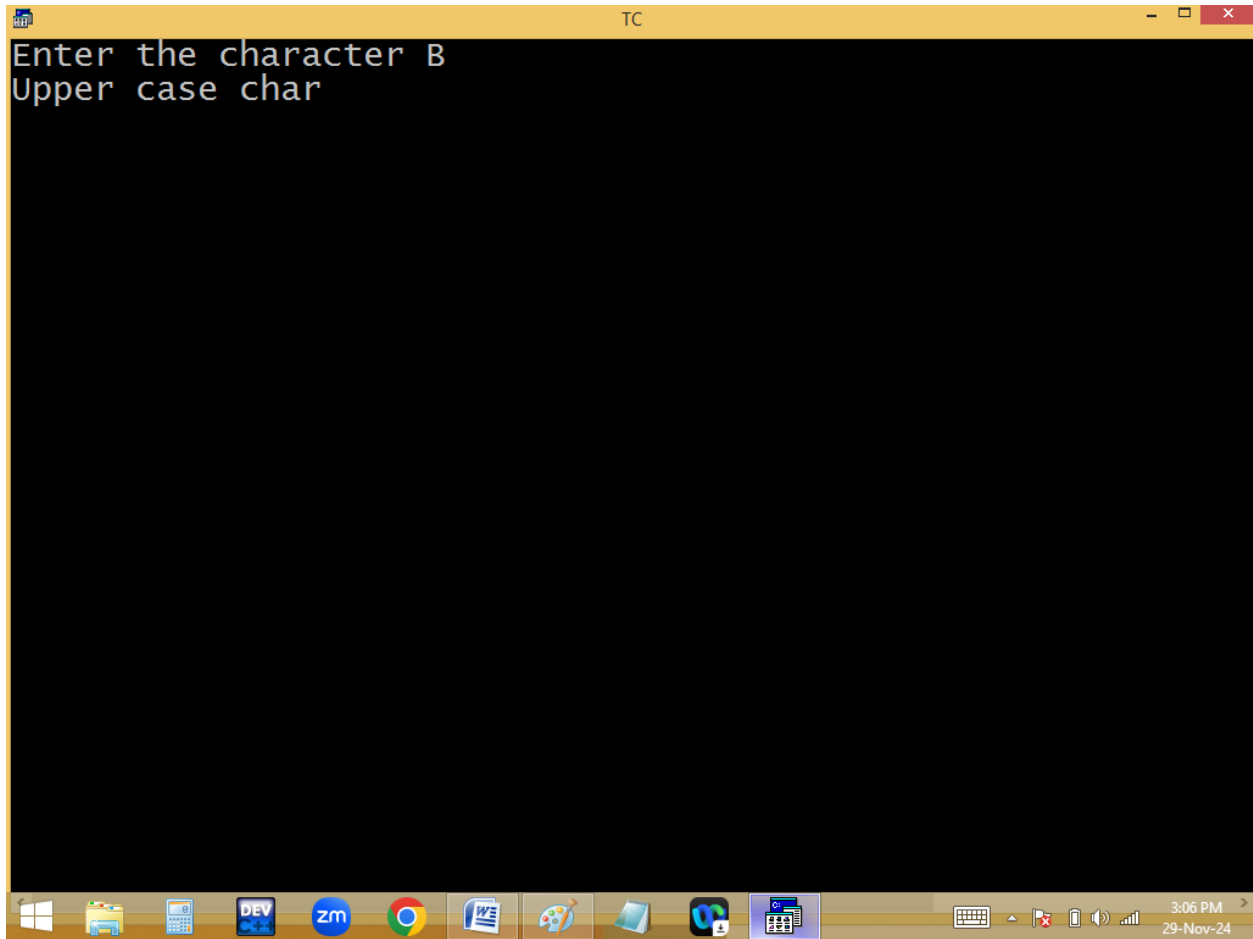
Finding upper or not?

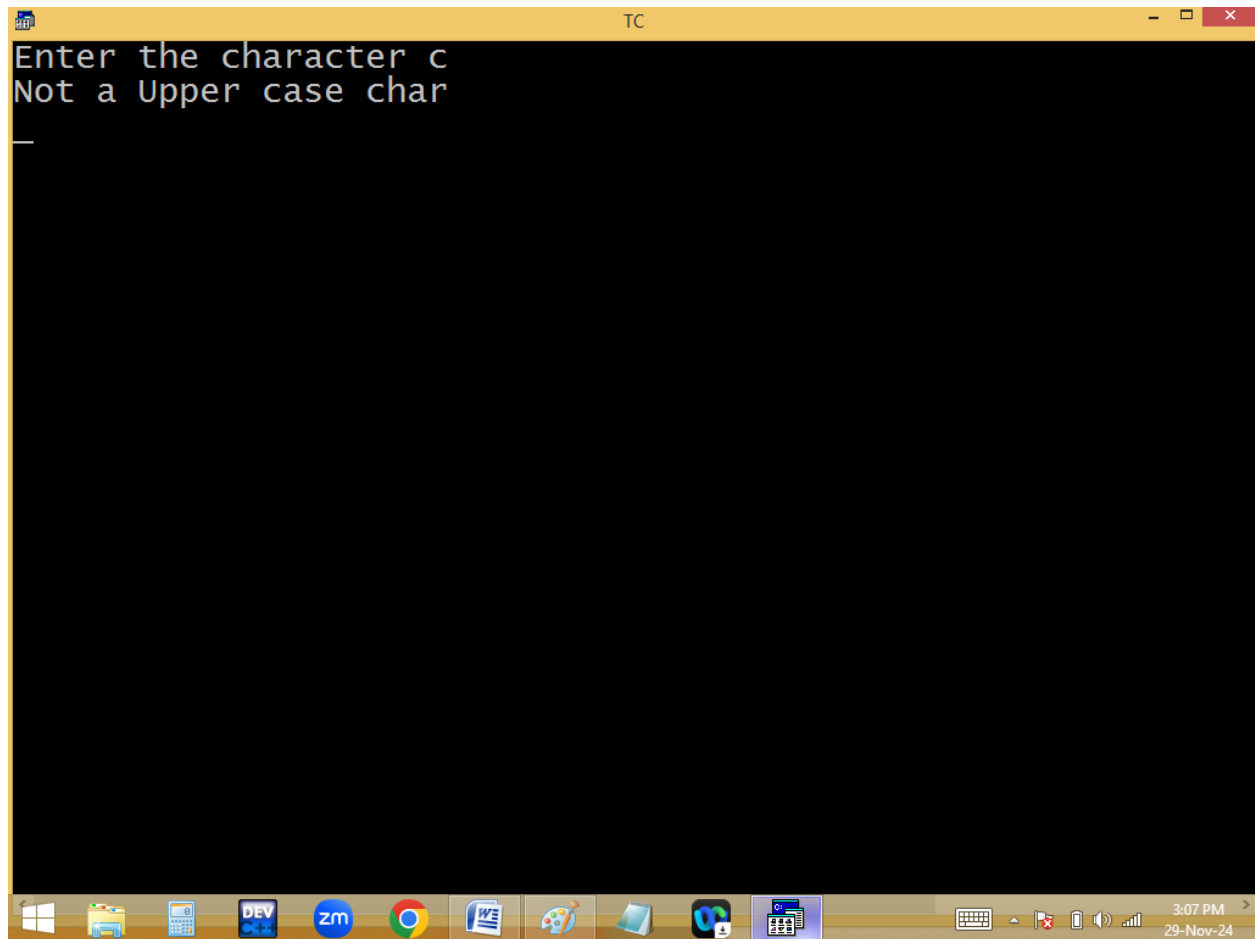


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below it is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Run" menu, there is a submenu with the following options: Line 9, Col 21, Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a dark blue background with yellow text. The code is as follows:

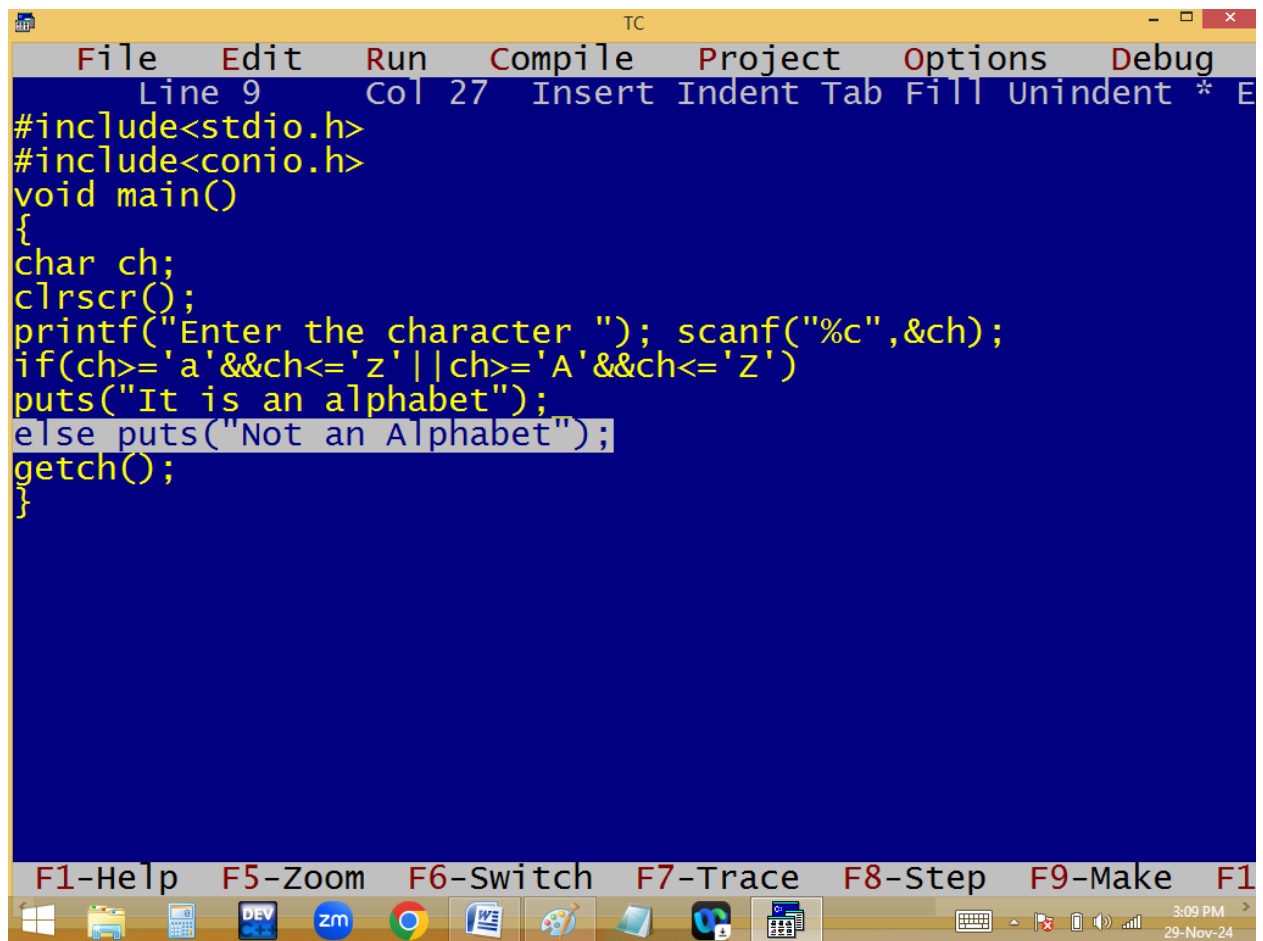
```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>=65&&ch<=90)puts("Upper case char");
else puts("Not a Upper case char");
getch();
}
```

The line `else puts("Not a Upper case char");` is currently selected with a light blue highlight. At the bottom of the IDE window, there is a status bar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. Below the IDE window is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system tray on the right shows the time as 3:06 PM and the date as 29-Nov-24.





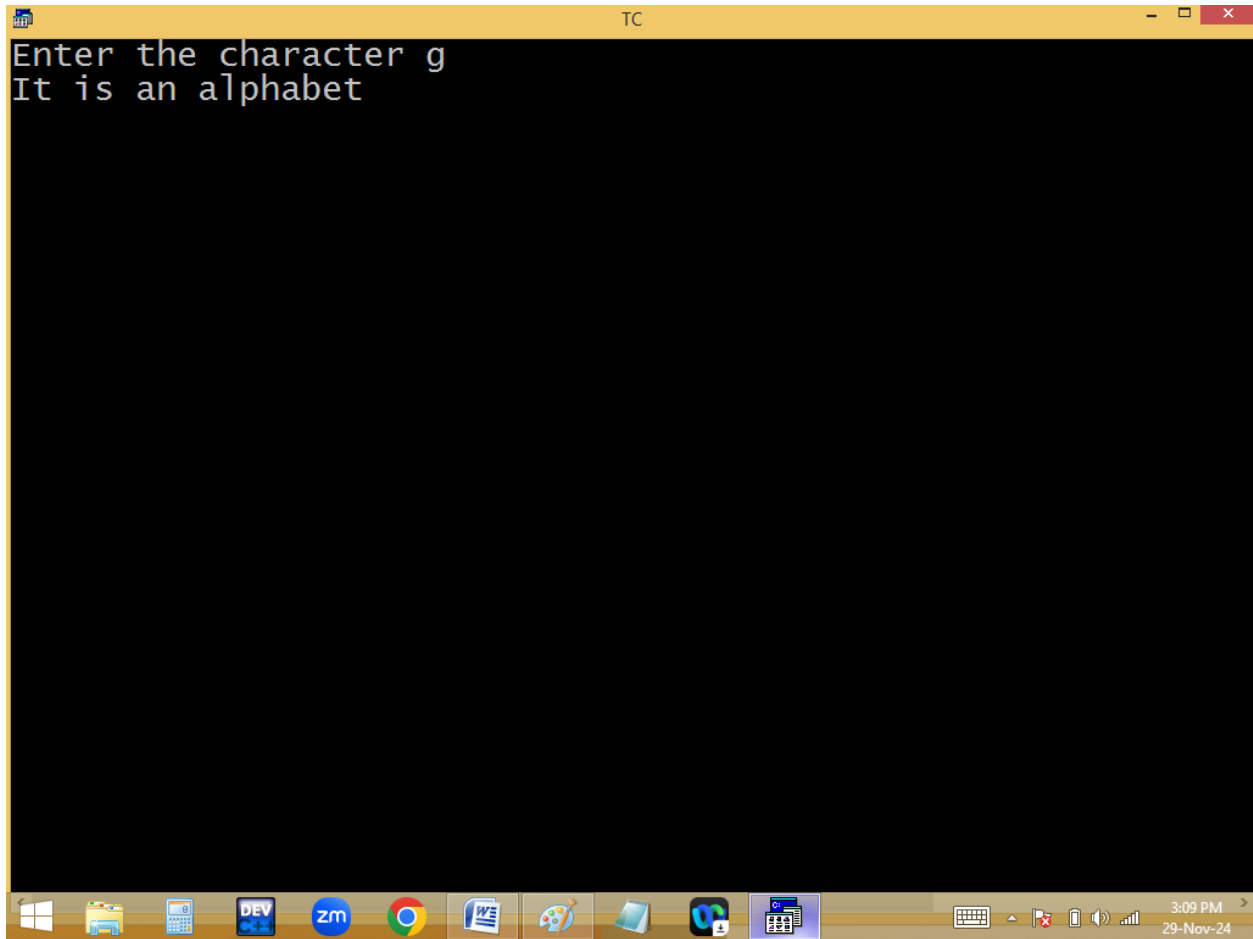
Finding alphabet or not?

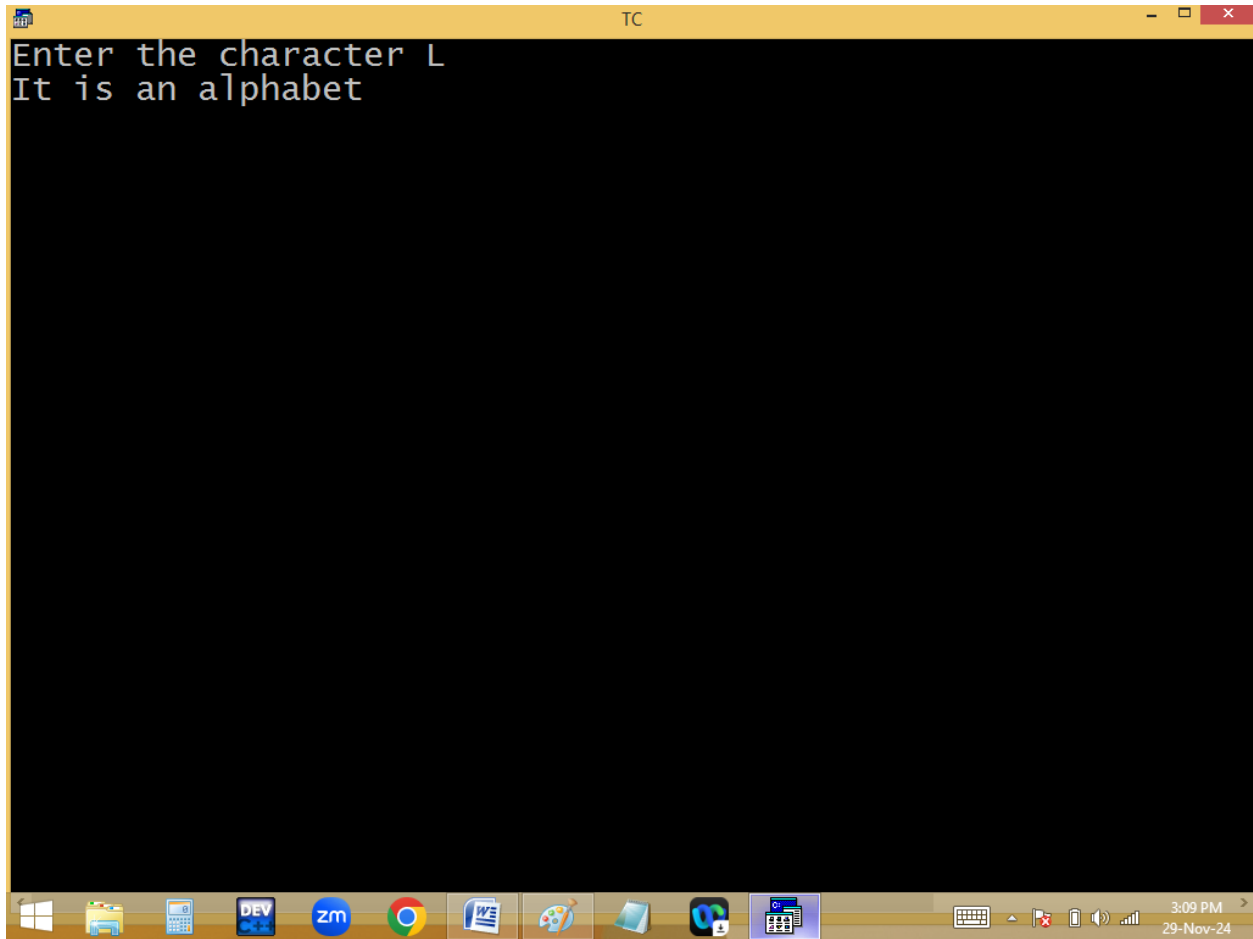


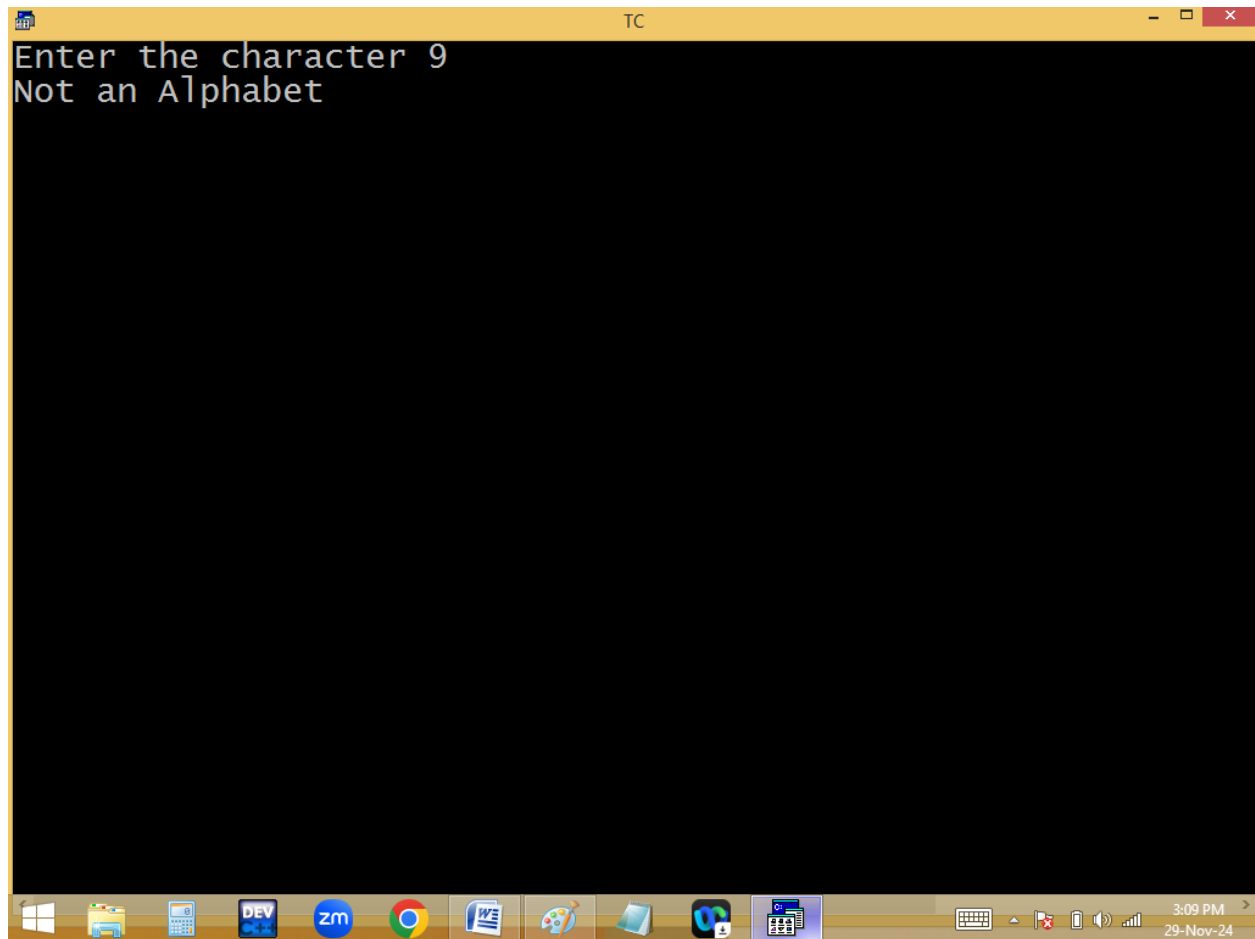
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 9", "Col 27", and "Insert Indent Tab Fill Unindent * E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a'&&ch<='z' || ch>='A'&&ch<='Z')
puts("It is an alphabet");
else puts("Not an Alphabet");
getch();
}
```

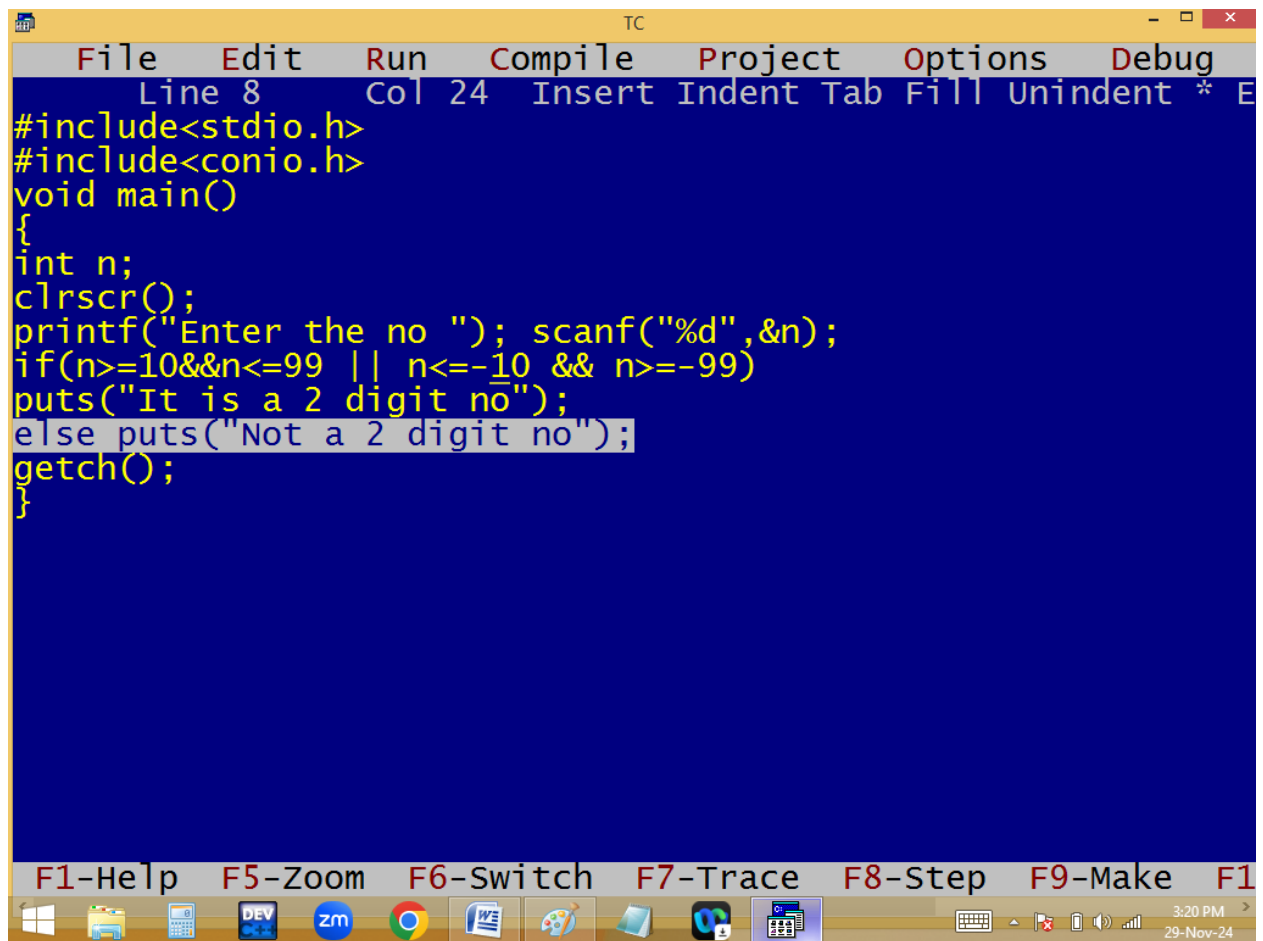
The line `else puts("Not an Alphabet");` is highlighted in light blue. At the bottom of the IDE, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the toolbar is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and a folder icon. The system tray on the right shows the time "3:09 PM" and the date "29-Nov-24".







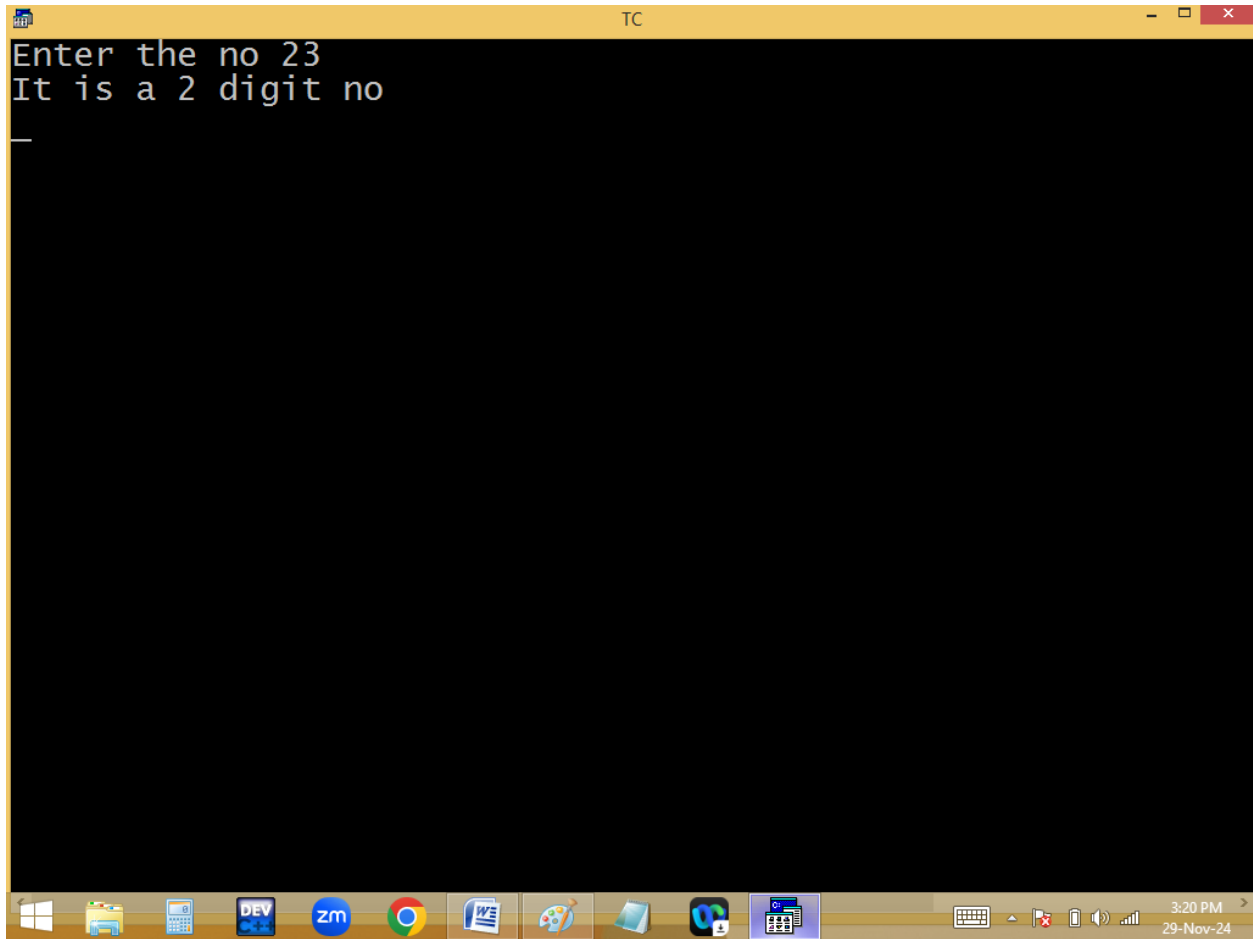
Finding 2 digit no or not?

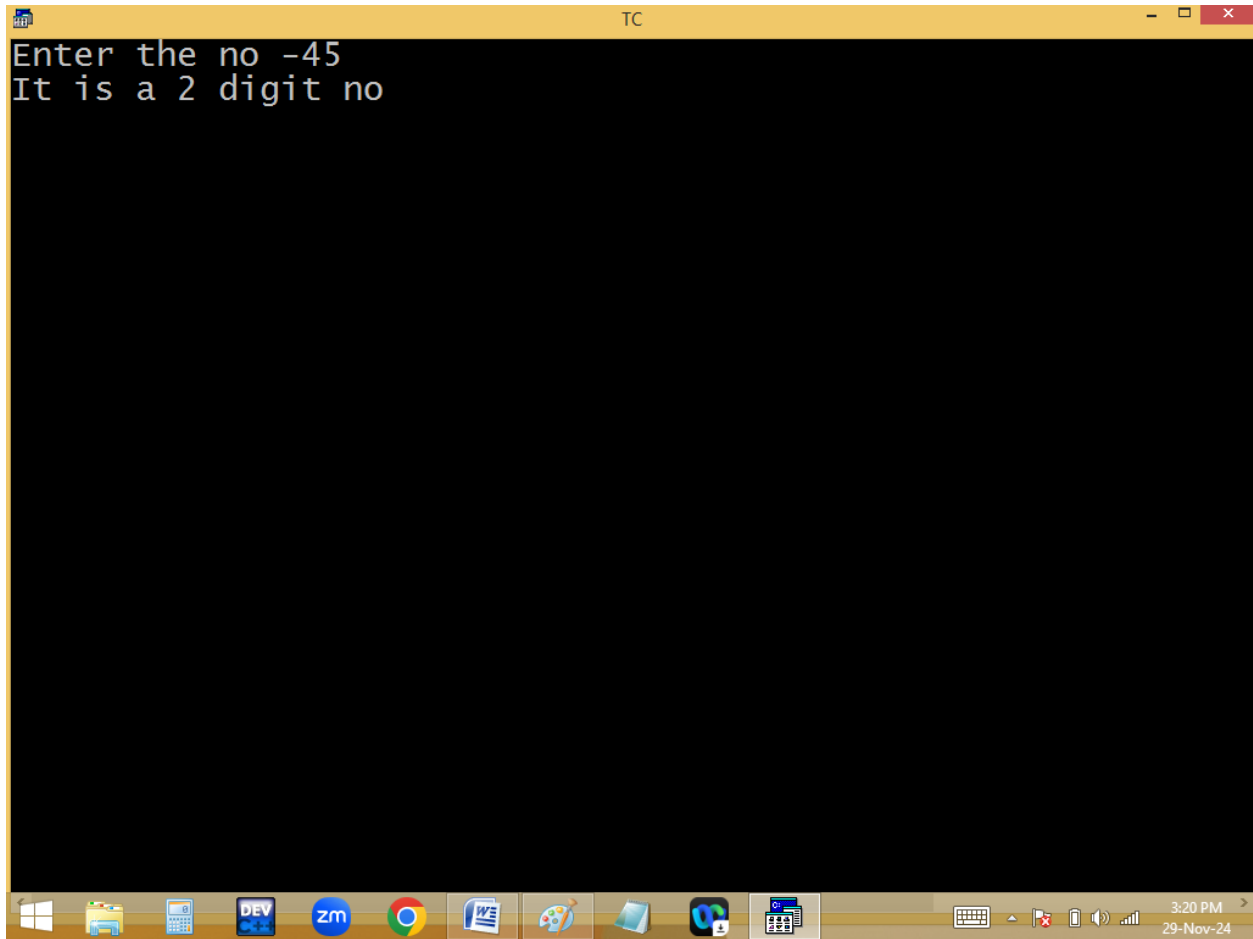


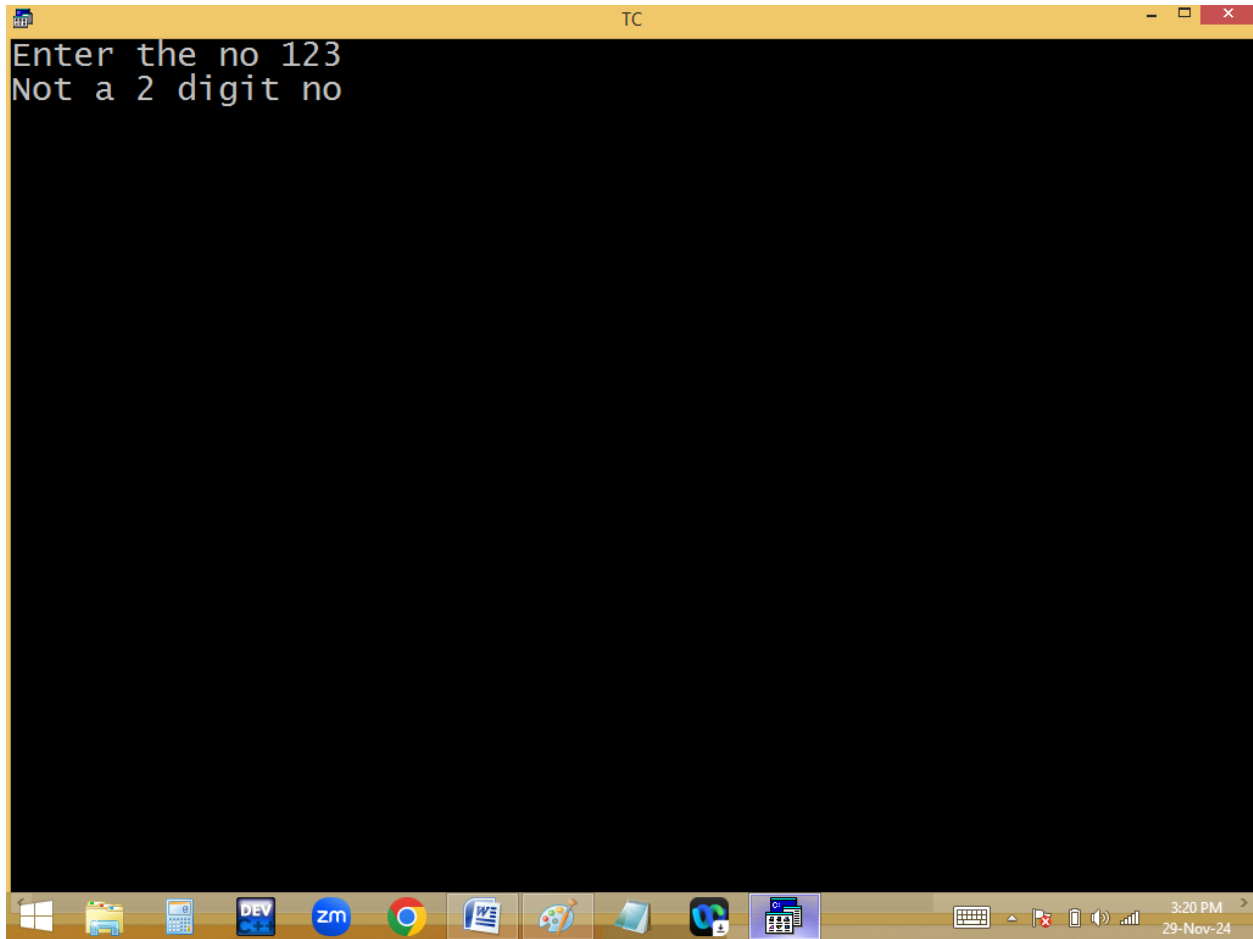
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 8", "Col 24", and "Insert Indent Tab Fill Unindent * E". The main editing area has a dark blue background with yellow text. The code is as follows:

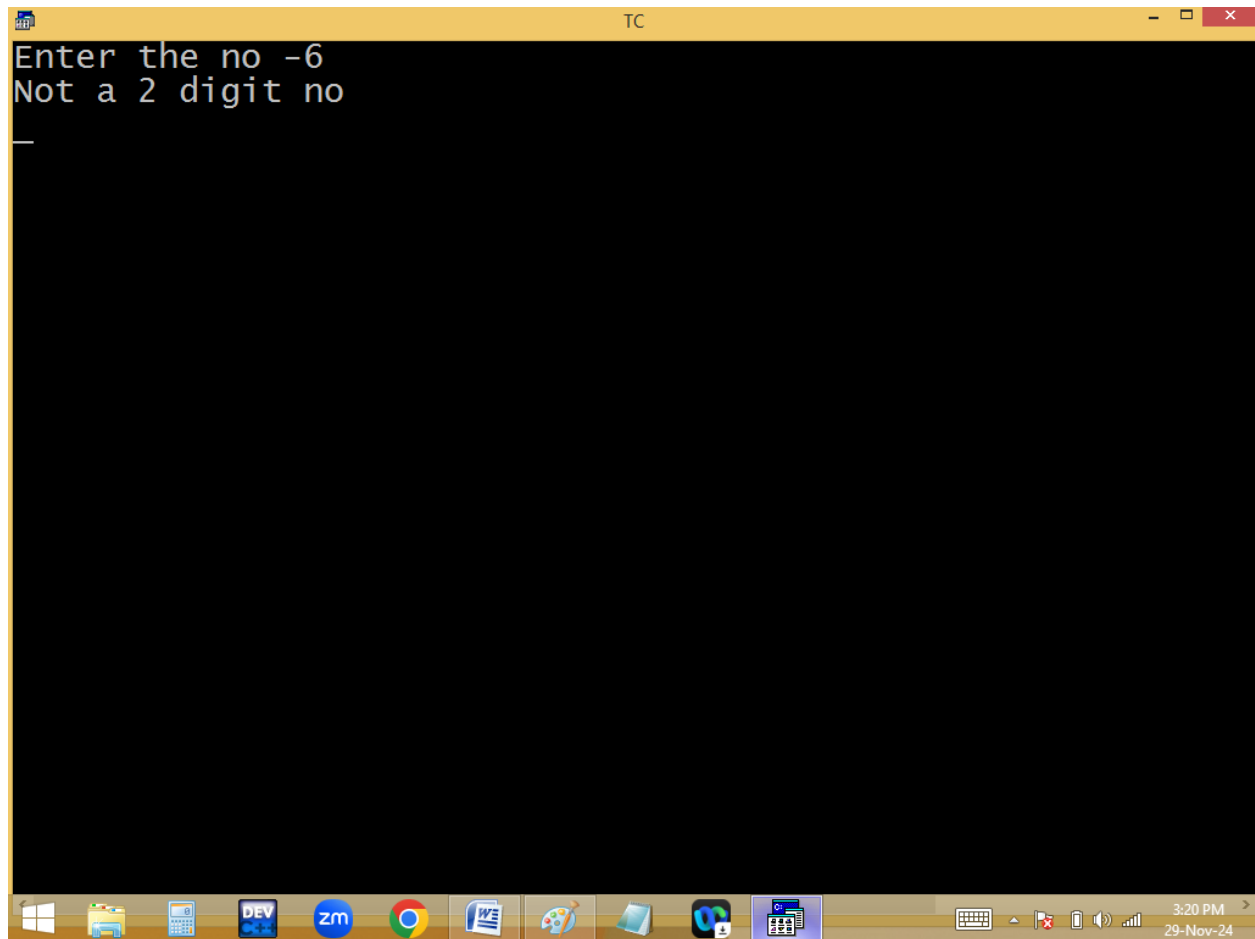
```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter the no "); scanf("%d",&n);
if(n>=10&&n<=99 || n<=-10 && n>=-99)
puts("It is a 2 digit no");
else puts("Not a 2 digit no");
getch();
}
```

The bottom of the window features a toolbar with icons for various functions and a status bar. The status bar displays the time "3:20 PM" and the date "29-Nov-24".

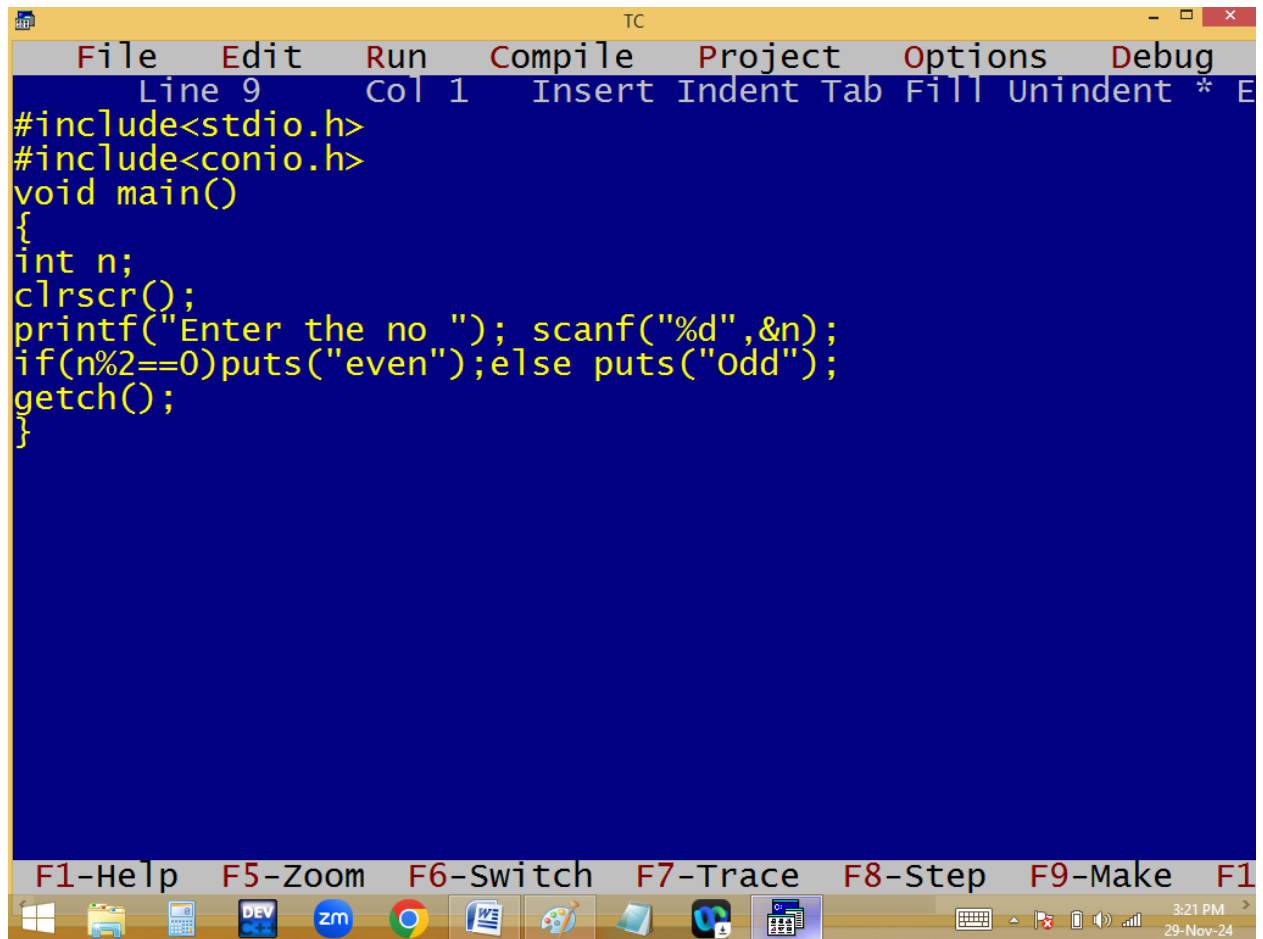








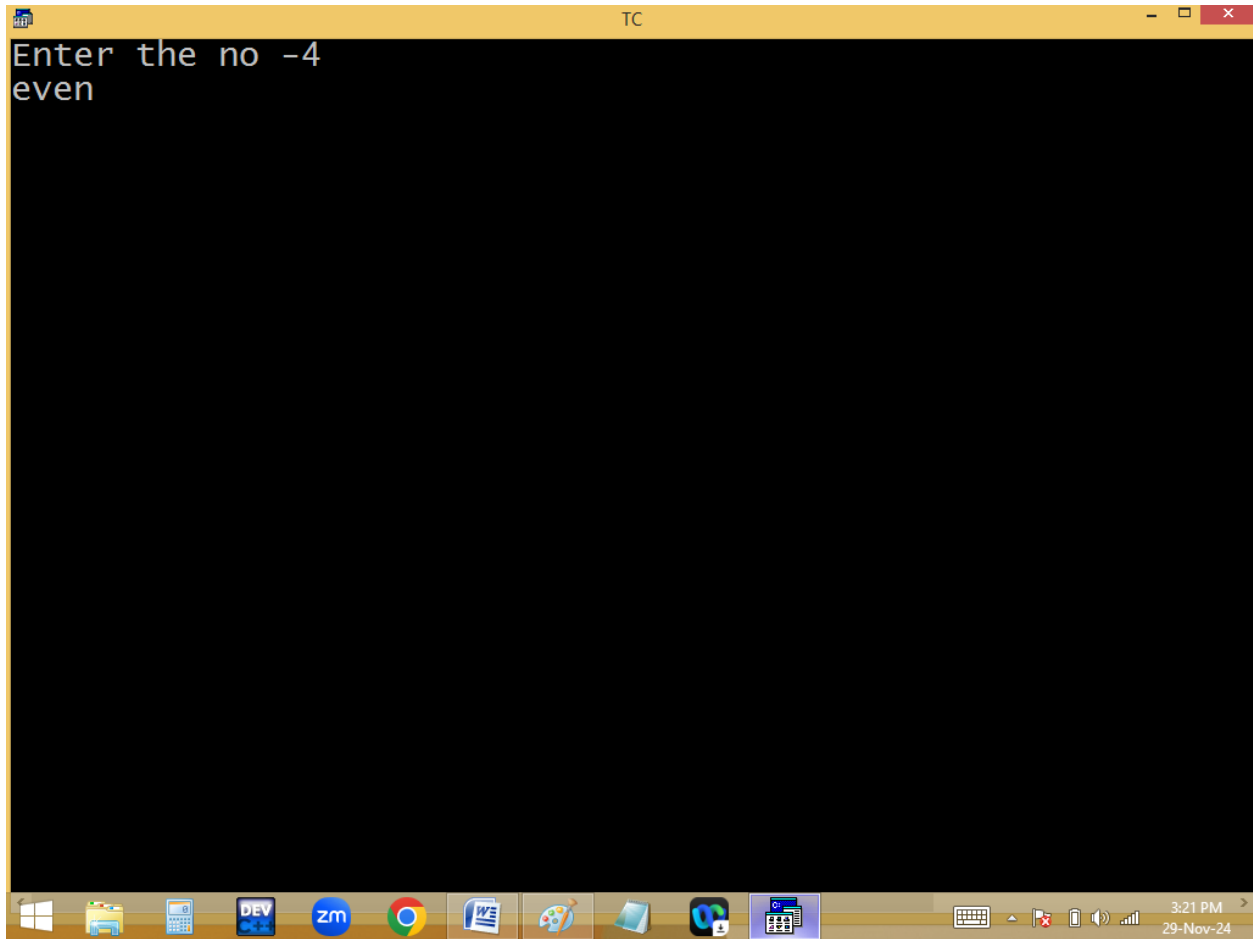
Finding even/odd:

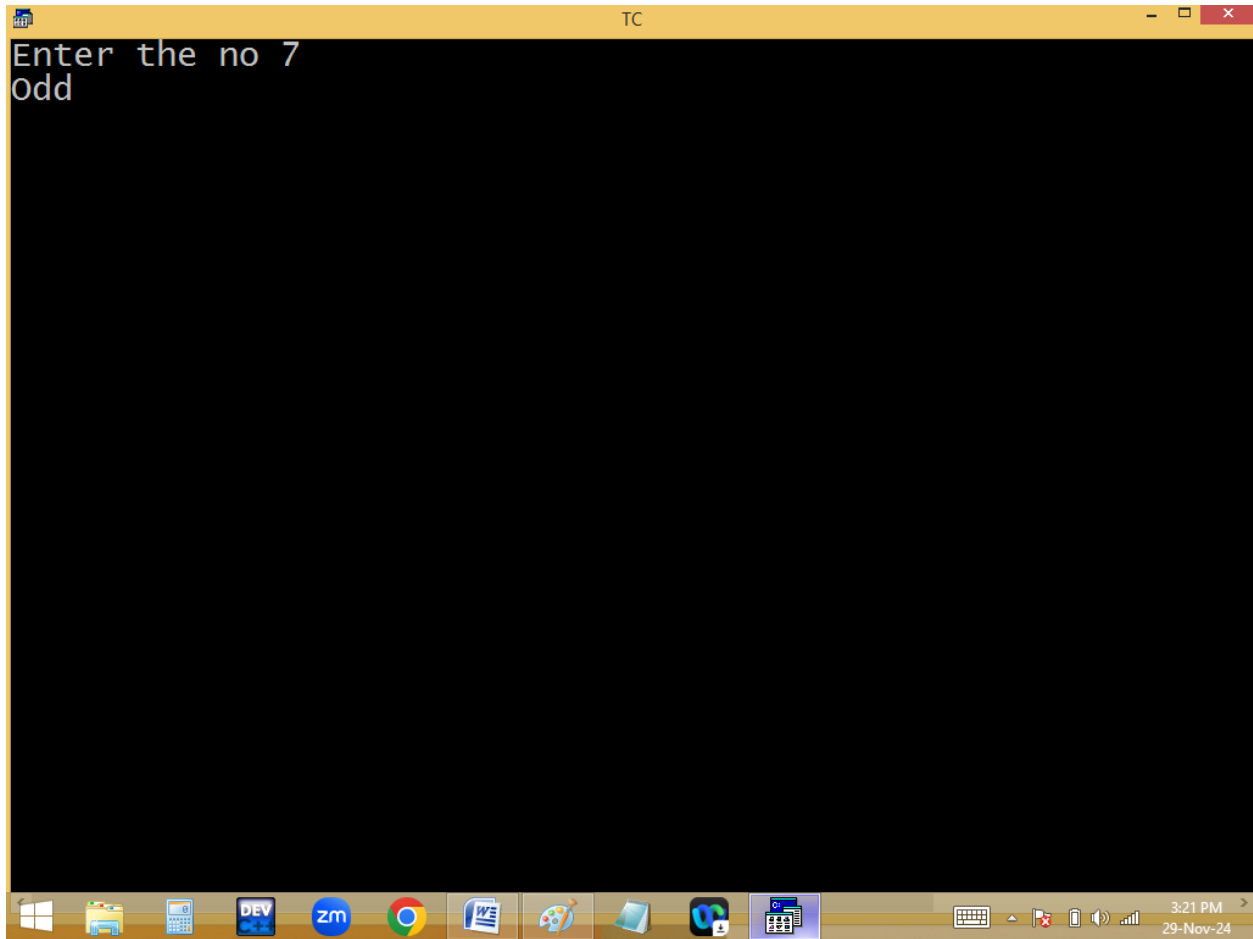


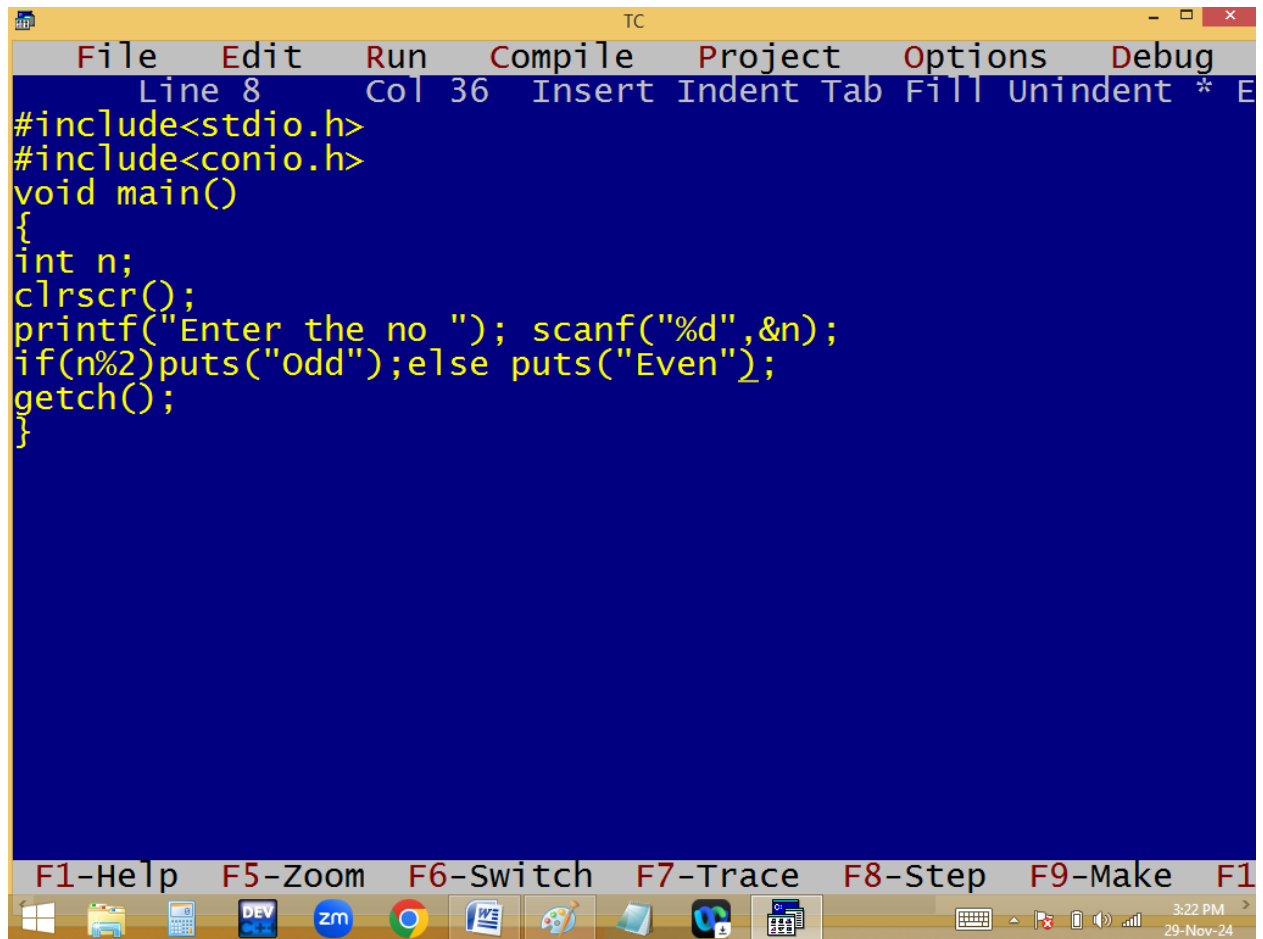
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 9", "Col 1", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter the no "); scanf("%d",&n);
if(n%2==0)puts("even");else puts("Odd");
getch();
}
```

At the bottom of the IDE window, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the toolbar is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder icon. The system tray on the right shows the time "3:21 PM" and the date "29-Nov-24".



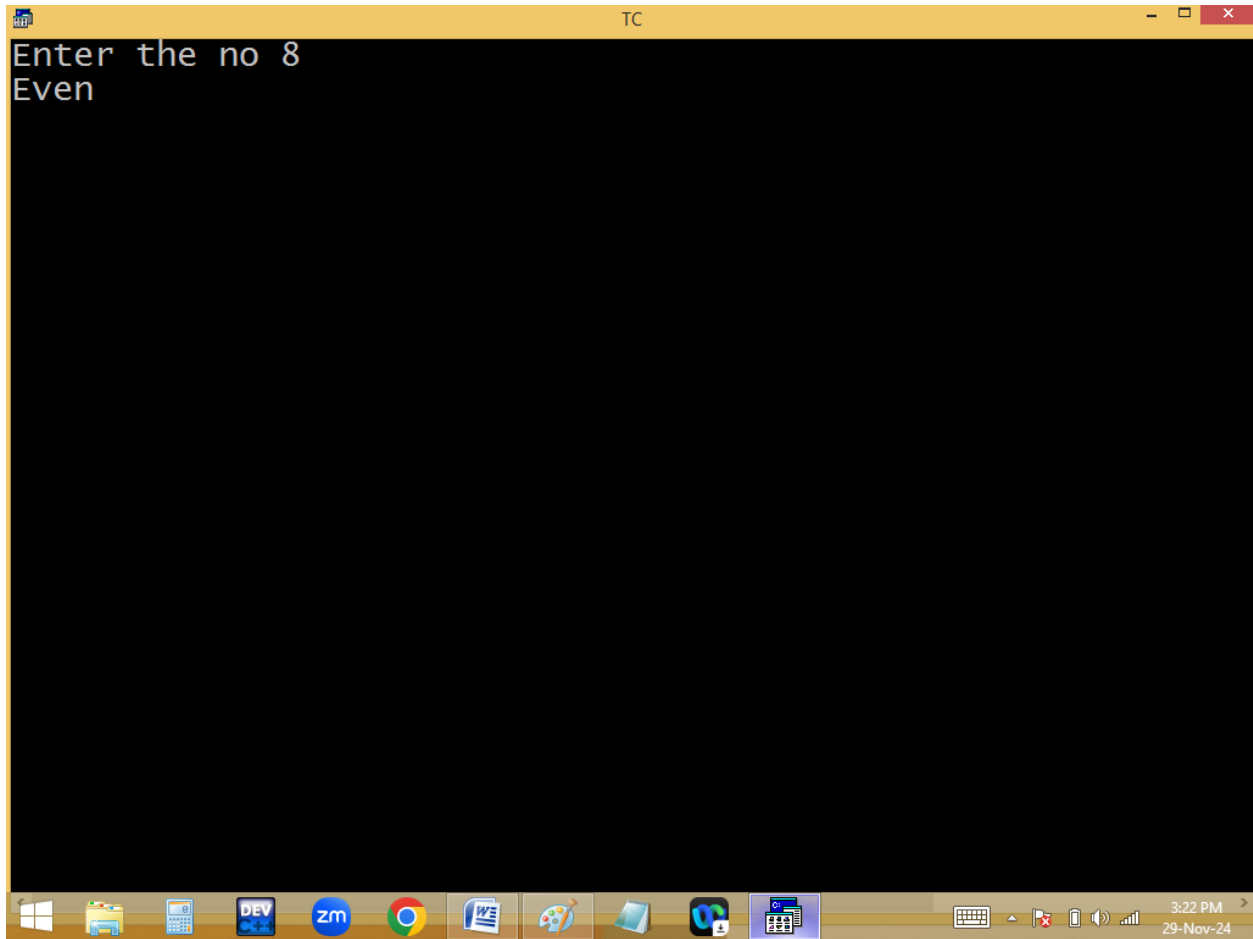


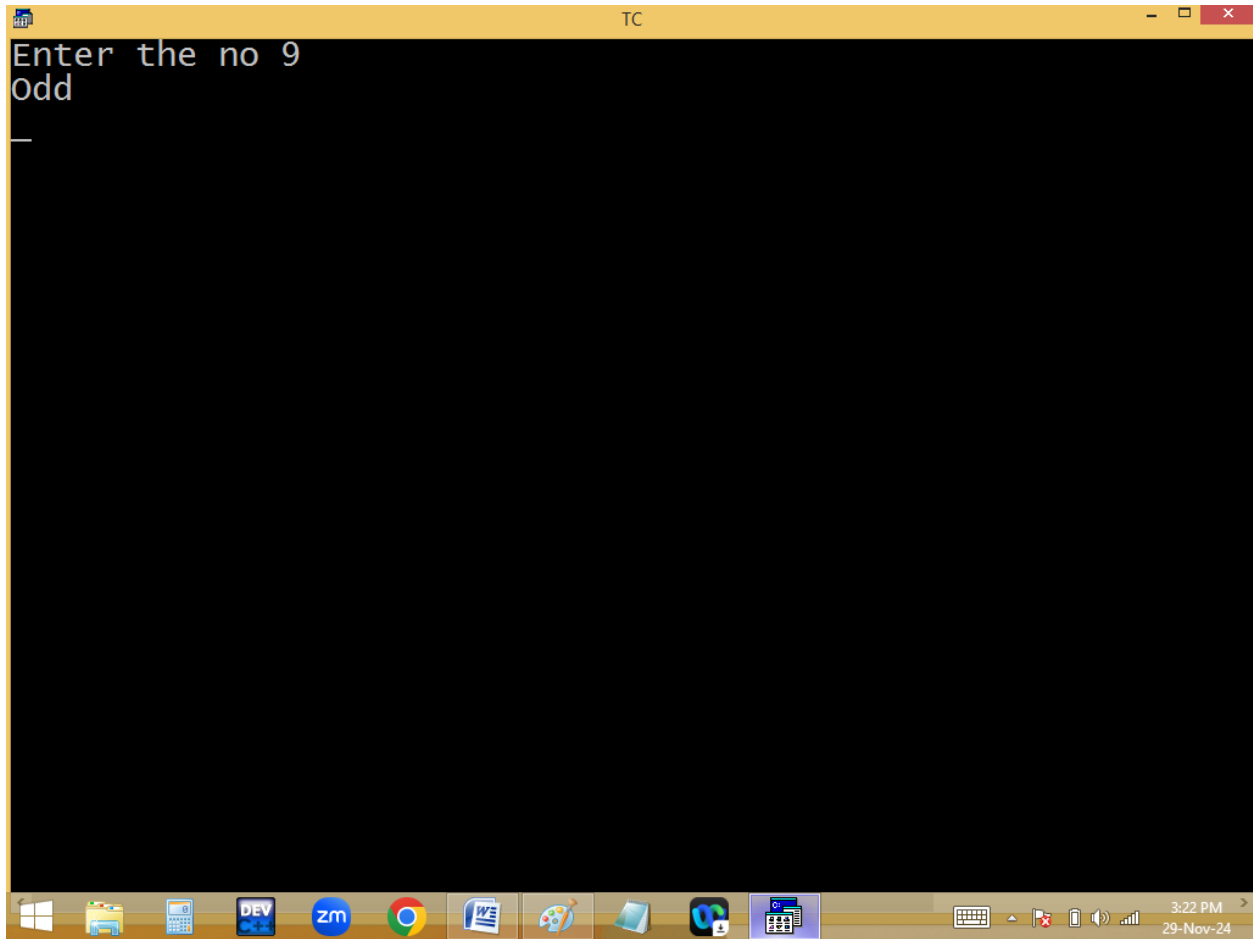


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 8", "Col 36", and a list of editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a blue background and contains the following C code:

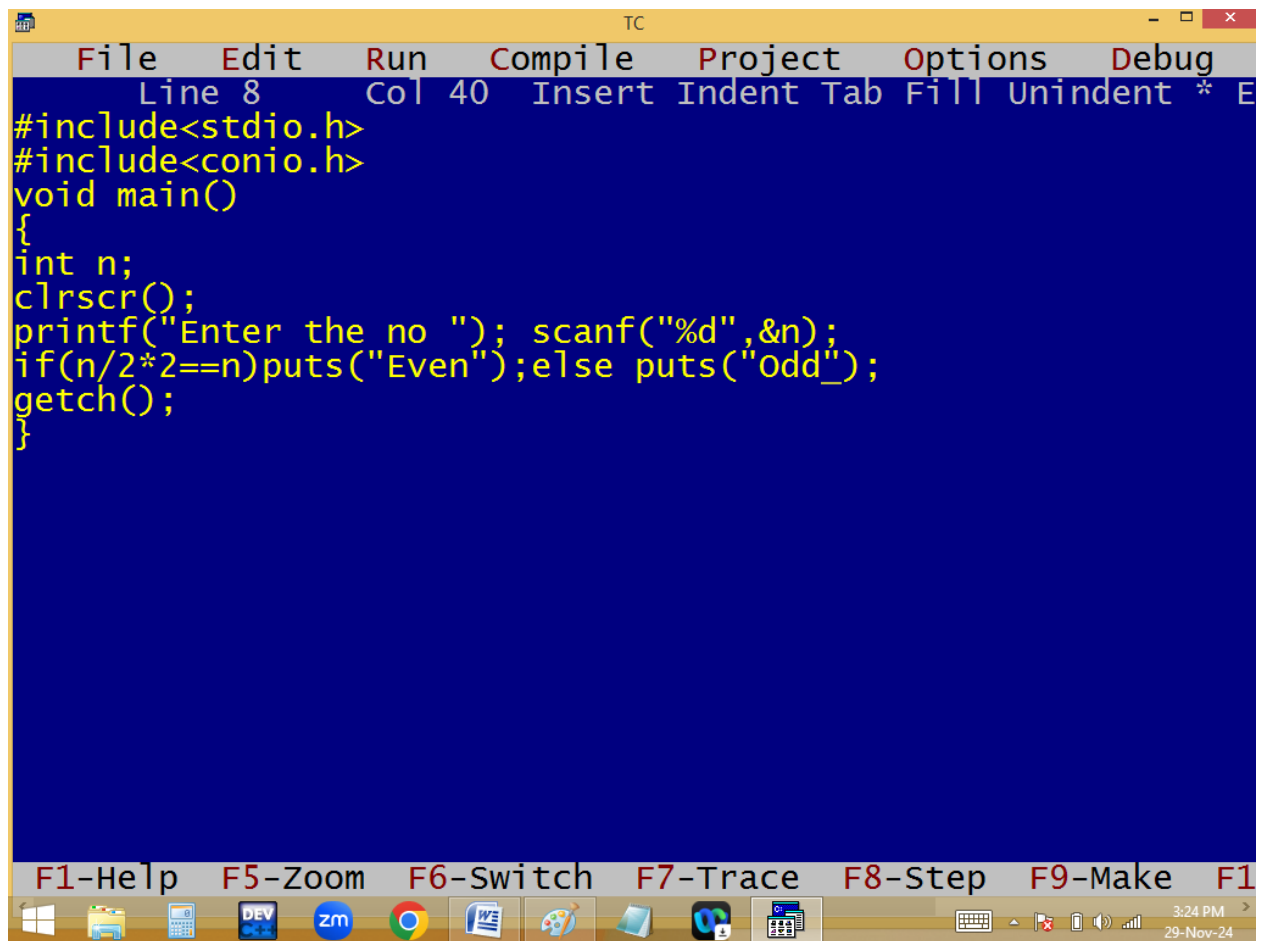
```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter the no "); scanf("%d",&n);
if(n%2)puts("Odd");else puts("Even");
getch();
}
```

At the bottom of the IDE window, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the IDE window is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and several other applications. The system tray on the right shows the time as "3:22 PM" and the date as "29-Nov-24".





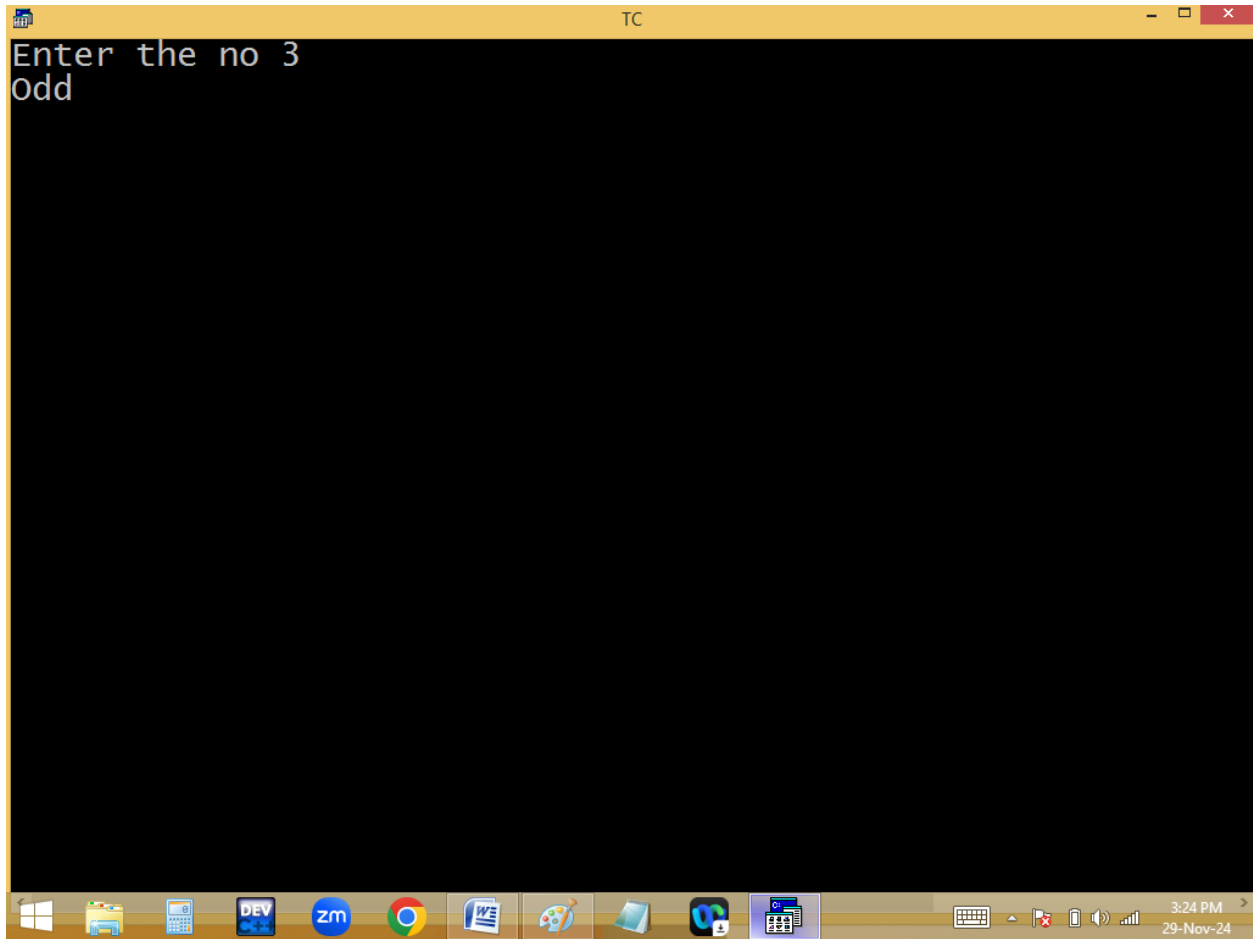
Without using %:

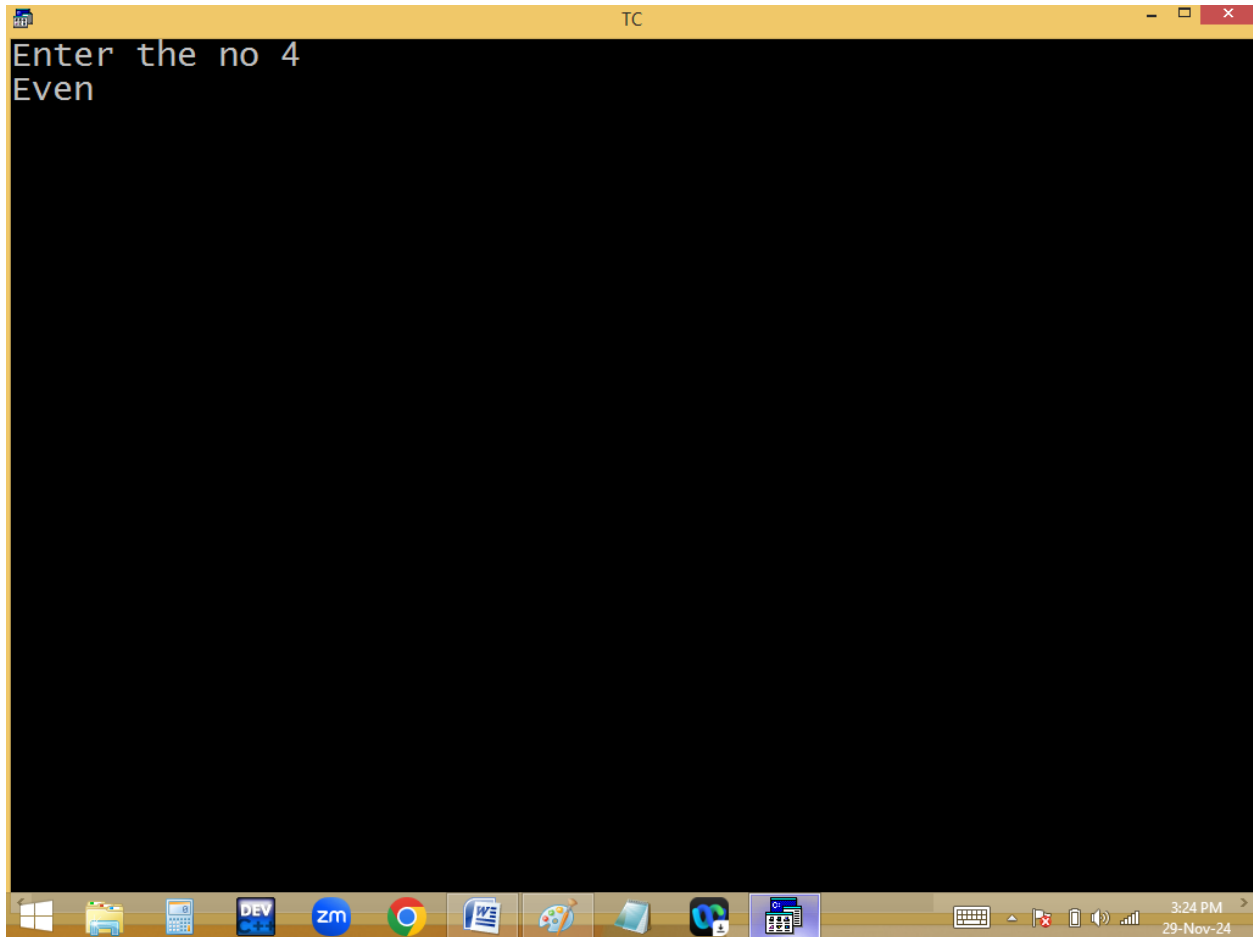


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 8", "Col 40", and various editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter the no "); scanf("%d",&n);
if(n/2*2==n)puts("Even");else puts("Odd_");
getch();
}
```

At the bottom of the IDE window, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the toolbar is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and a folder icon. The system tray on the right shows the time "3:24 PM" and the date "29-Nov-24".

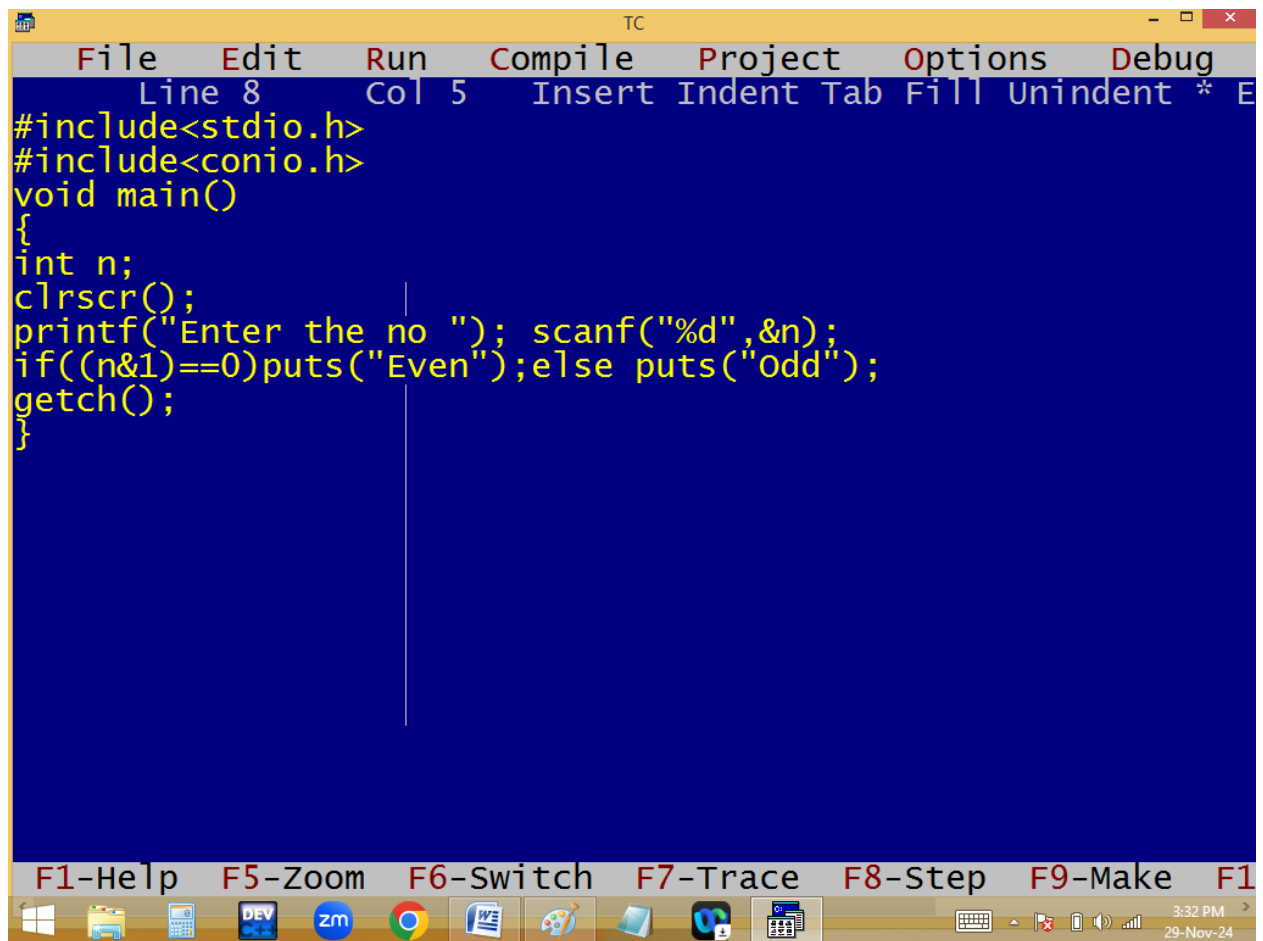




$4/2=2*2=4==4$ true
if ($n/2*2==n$) p(even); else p(odd);

$5/2=2*2=4==5$ false

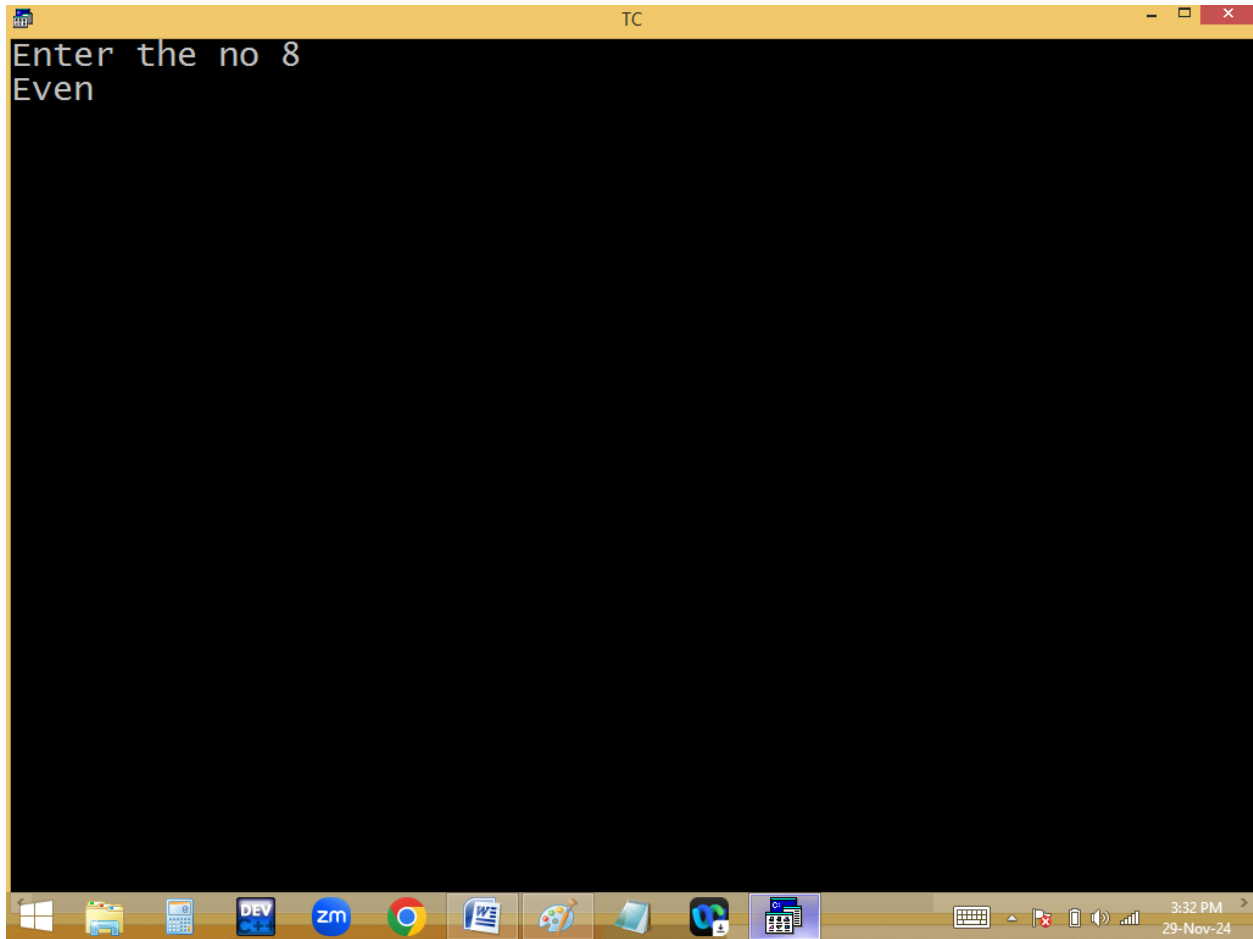
Without using arithmetic operators:

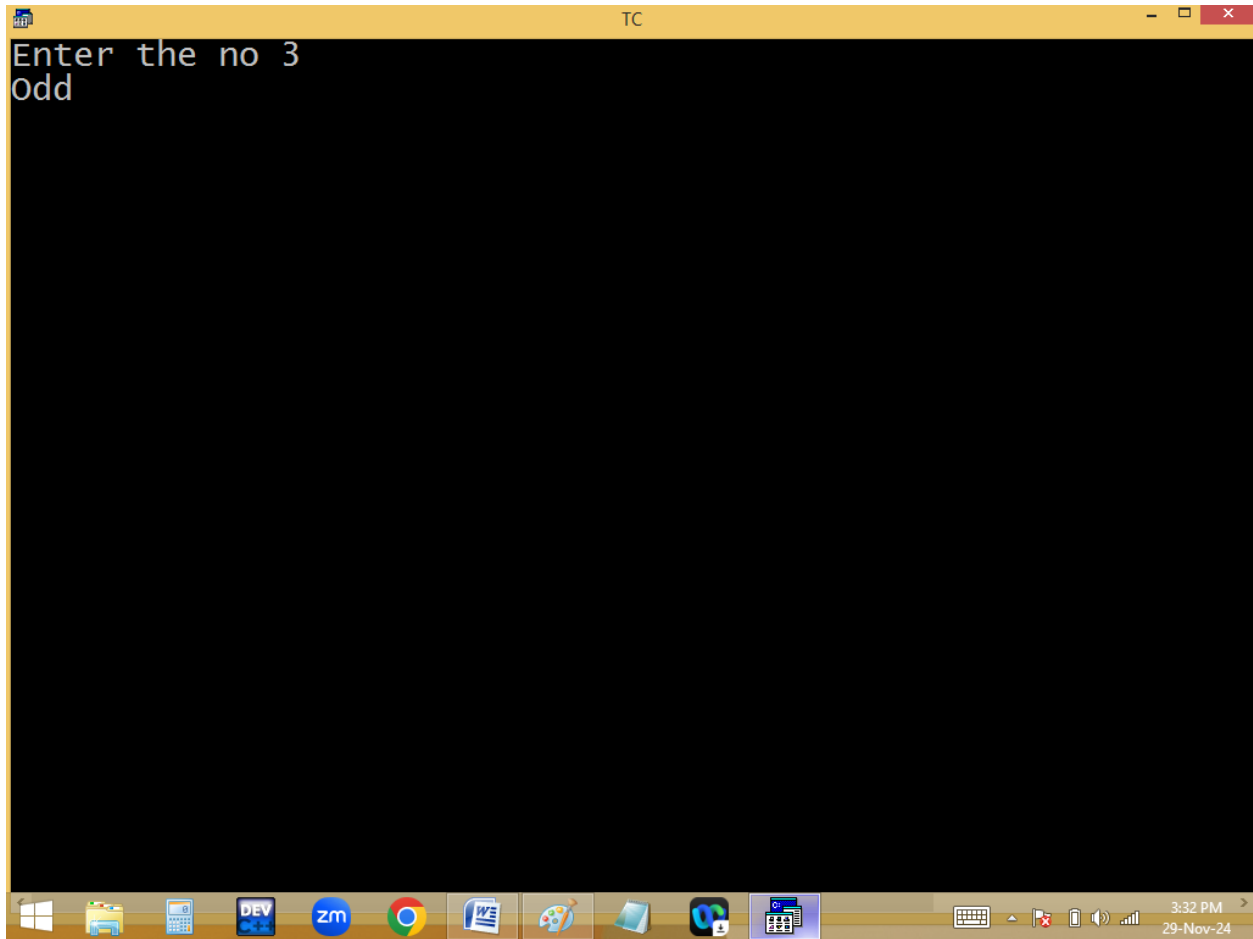


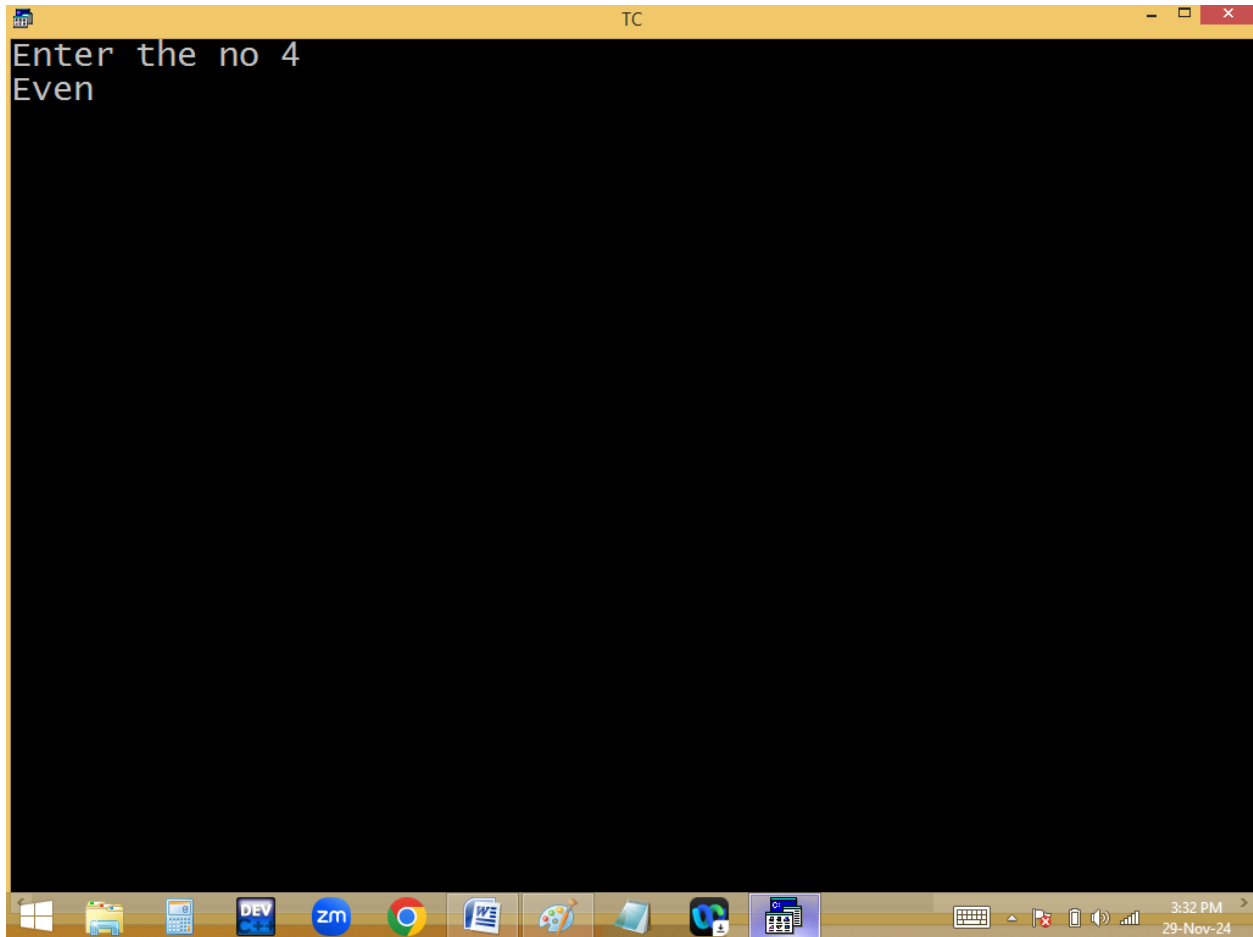
The image shows a screenshot of the Turbo C++ (TC) integrated development environment. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 8", "Col 5", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter the no "); scanf("%d",&n);
if((n&1)==0)puts("Even");else puts("Odd");
getch();
}
```

At the bottom of the window, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other applications. The system clock in the bottom right corner indicates the time is 3:32 PM on 29-Nov-24.







if(3 & 1 == 0)p(even);else p(odd);

3 = 11
1 = 01
01=1

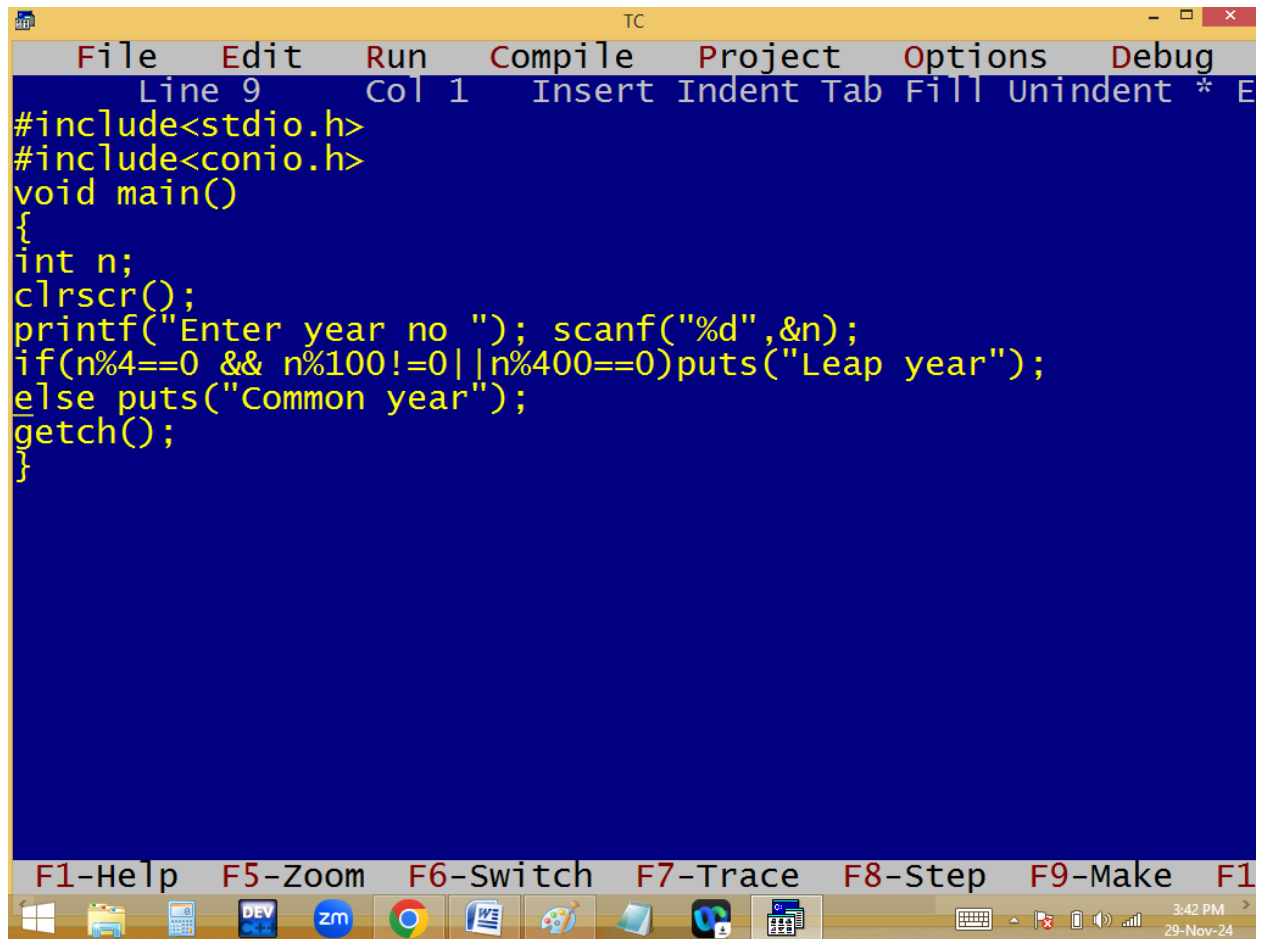
if(4 & 1 == 0)p(even);else p(odd);

4 = 100
1 = 001
000=0

2 | 3
1-1

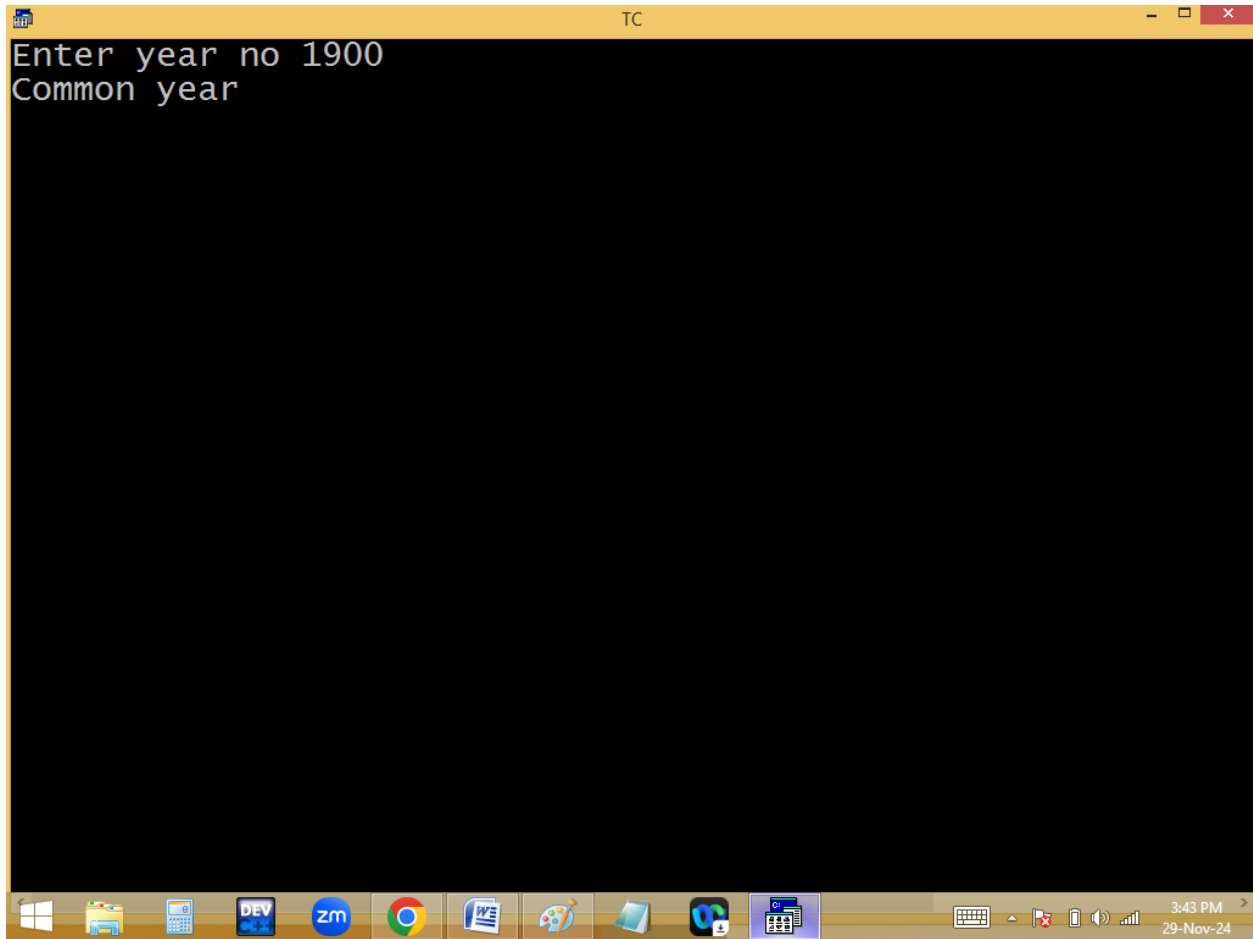
2 | 4
2 | 2-0
1-0

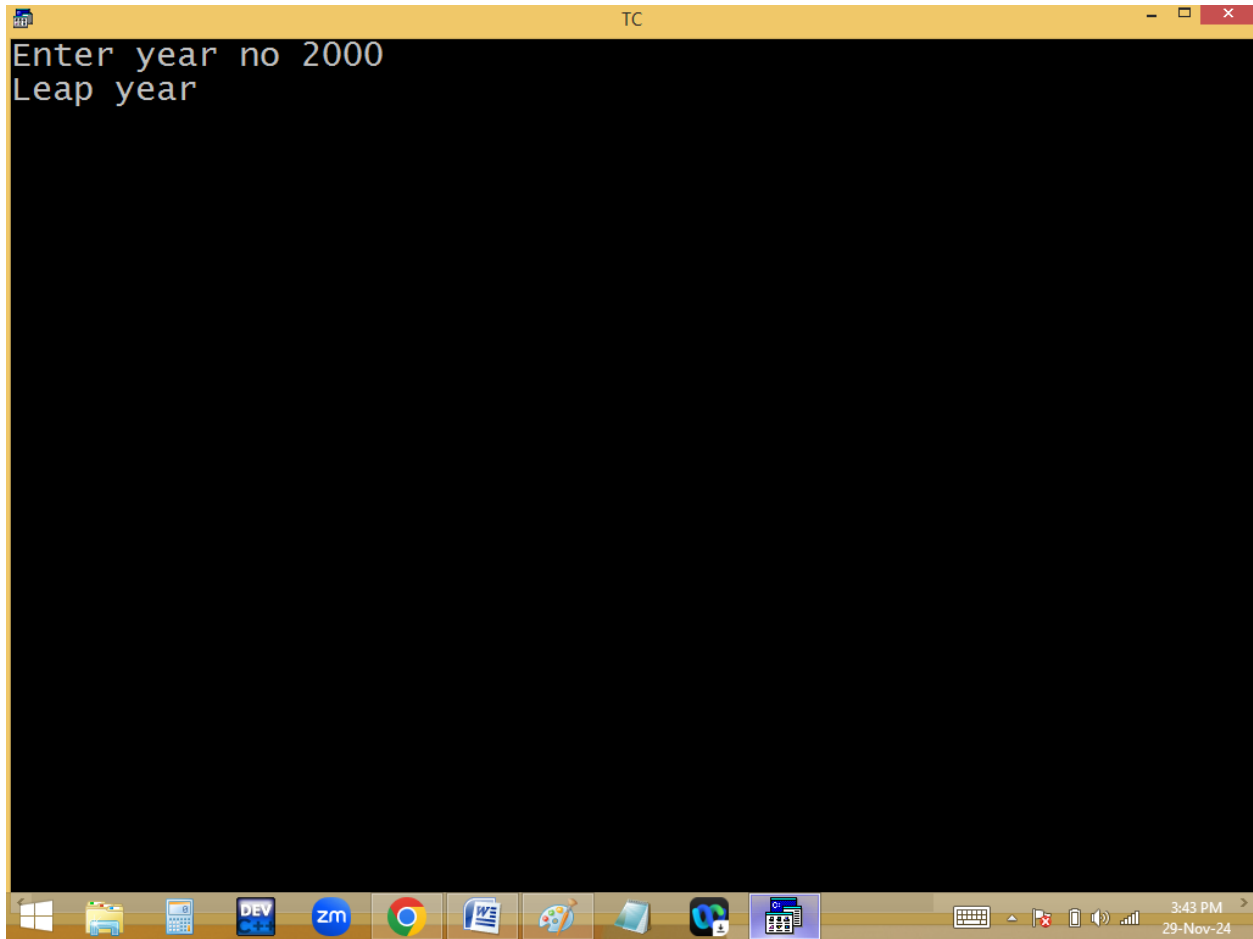
Finding leap / common year:

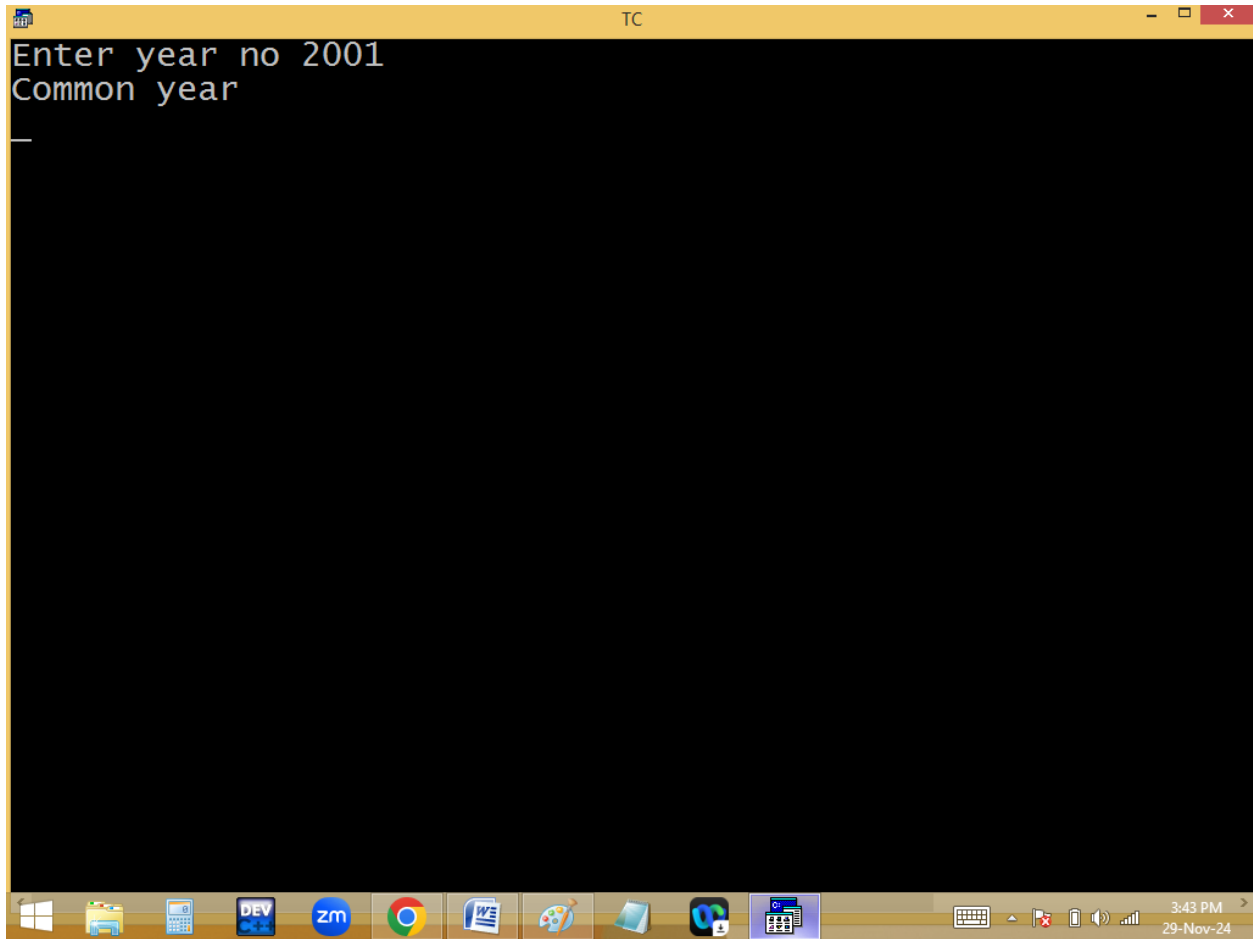


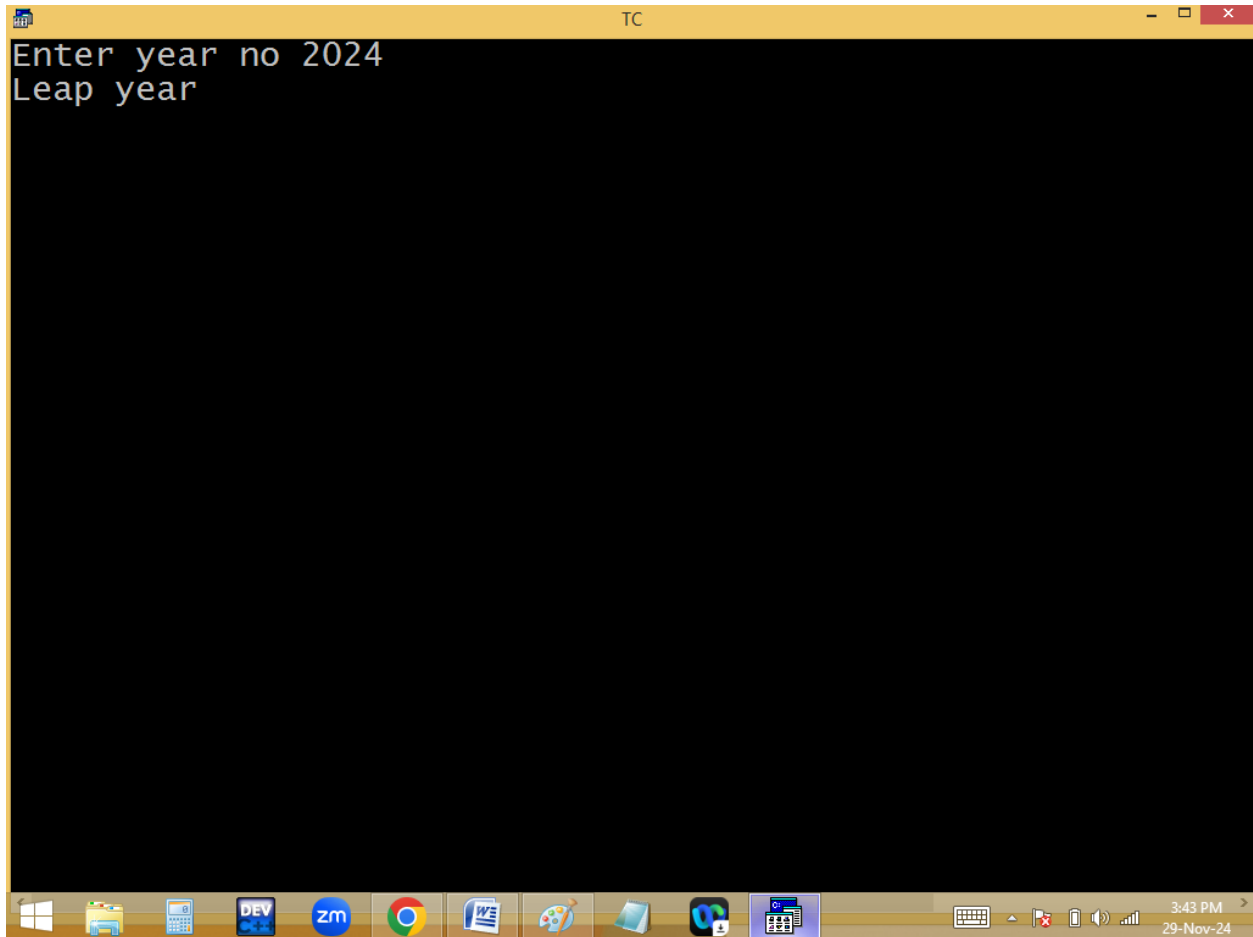
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status line indicates "Line 9 Col 1" and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and a help key. The main editing area has a dark blue background with yellow text. The code is a C program to check if a year is a leap year. It includes `<stdio.h>` and `<conio.h>`, defines a `main()` function, declares an integer `n`, clears the screen with `clrscr()`, prompts the user to enter a year, and uses `scanf` to read the input. It then checks the leap year condition: `if(n%4==0 && n%100!=0 || n%400==0)`. If true, it prints "Leap year"; otherwise, it prints "Common year". Finally, it calls `getch()` to pause the program. The bottom of the window features a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar at the very bottom shows various application icons and the system clock indicating 3:42 PM on 29-Nov-24.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter year no "); scanf("%d",&n);
if(n%4==0 && n%100!=0 || n%400==0)puts("Leap year");
else puts("Common year");
getch();
}
```









Find the sales man gross salary using below table:

Sales$\geq$100000	Sales$<$100000
Basic=10000	Basic=10000
Da=65.5% of basic	Da=65.5% of basic
Bonus=10000	Bonus=2000
Commission=10% of sales	Commission=5% of sales
Gross salary = basic + da + bonus + commission	